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Theories and definitions of the informal economy: A survey

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Abstract

This paper surveys the literature on the definitions and theories of the informal economy (IE). It outlines the main differences in the definitional approaches of informality and proposes a definition for the IE that may be consistent with measurement and policy. The literature review distinguishes three, not mutually exclusive, approaches to the IE: the “neoclassical,” the “macro-econometric,” and the “conceptual” perspectives. I conclude the survey by outlining a framework that points out the main differences and similarities among the schools of thought on the nature of the IE. From a normative perspective, a leading recommendation emerges: a “one-size-fits-all” policy response is inadequate to deal with the heterogeneous nature of the IE. Policy measures should be tailored around the distinct segments of the IE grounded in terms of characteristics, contexts, and needs of the informal units.

KEYWORDS

informal economy, informality, informal sector, shadow economy

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1 | INTRODUCTION

Informal economy (IE) encompasses a huge diversity of phenomena that vary in composition and size within economies and across regions. Although estimating the size of informality is challenging, there is a wide consensus that the IE is the most important source of employment in developing countries and it also has a significant role in providing employment and income for marginalized people in developed economies. Specifically, according to the International Labour Organization (ILO, 2018a), 61.2% of global employment (about two billion people)¹ is informal.

Literature on informality dates back to the analyses of the British anthropologist Keith Hart, who coined the term “informal sector”² in 1971 publishing it 2 years later in his article (Hart, 1973) and in the ILO’s (1972) Report on Kenya, which contributed to the widespread use of this “new” concept. Hart’s most important (1973: 61) contribution to this literature consisted in showing that the economic activities of the urban Southern Ghanaian sub-proletariat were not a passive and exploited “*reserve army of urban unemployed and underemployed*” but possessed “*some autonomous capacity for generating growth in the incomes of the urban (and rural) poor.*” As Clement (2015) points out, this perspective disagreed with the prevailing theory of economic development at that time (often labeled “Modernization” theory), which regarded the “informal sector” as a transient phase in the process of economic development. Indeed, while the “Modernization” theory assumed subordination between the traditional/subsistence (i.e., informal) and the modern/industrialized (i.e., formal) sectors, for Hart (1973) and the ILO (1972), the informal sector was too large, persistent and, in some places procyclical, to predict its natural disappearing with economic development. After these seminal studies, mainstream thinking progressively moved away from the “Modernization” theory of the IE. The idea that there was only one, inevitable trajectory of development, which corresponded to the formalization of informal activities into the official economy, became just one of the potential theoretical views. Following this strand of research, several scholars have classified theories on informality but, at the state of art, these taxonomies are sometimes inconsistent with each other. For instance, we can find different labels for the same theoretical view as well as the same label to indicate diverse theoretical approaches.

In this paper, I review the approaches followed in economic literature to explore the phenomenon of informality and, the schools of thought proposed to capture the intrinsic nature of the IE. As a result, I propose a unified conceptual framework of the main hypotheses behind these theories that may be useful to further foster an interdisciplinary debate on this issue.

In general, the vast literature on IE can be divided into three major (interconnected) issues: a definitional issue (i.e., what is “informal”); a measurement issue (i.e., “how” to estimate its size), and a theoretical explanation of “informality” (i.e., what are its causes and consequences? Why is it so persistent over time? What is its relationship with the formal economy?).

This paper deals with the first and third issues.³ Section 2 reviews the different definitional approaches to IE and, by pointing out the main critical issues, proposes a definition that aims to be consistent with statistical and economic purposes. Section 3 distinguishes three, partially overlapping, perspectives to approach IE in literature. Moreover, it outlines a conceptual framework to point out the main differences and similarities among the best-known theories of informality. The last section concludes with normative implications and some suggestions for further research.

2 | DEFINITION(S) OF IE

Hans Singer, who led the first ILO employment mission along with Richard Jolly in Kenya in 1972, compared the informal sector to a giraffe in that it is difficult to define by usual standards, but easy to recognize when you meet one. Lautier (1990) replied to Singer, by saying that the metaphor of the giraffe was inadequate. The informal sector is “*a unicorn because the literature abounds with definitions, but you will never have the opportunity to meet one, because it does not exist*” (Charmes, 2016, p.6). This disagreement on the best faunal metaphor for the informal sector gives us an idea of how challenging the debate on the definition of informality is.

An important source of misunderstandings on the definition of IE consists in the divergence of research goals among scholars from different disciplines (i.e., “statistical” vs. “economic”). For instance, the “economic” concept of IE (or of its seeming synonyms, e.g., shadow, unofficial, hidden, black, underground economy) has a different meaning to the concept of informality used by statisticians in national accounting (e.g., non-observed economy (NOE), underground production) or by labor economists in ILO statistics (e.g., informal sector and informal productive activities (IPA)).

Although there have been some attempts to reconcile these definitional approaches (e.g., Dell'Anno, 2007, 2016; Hussmanns, 2004; ILO, 2013, 2018a, 2021; OECD, 2002; Quiros-Romero et al., 2021), a common solution is still not available. For example, according to the strand of literature that focuses on informality from an “economic” perspective, Feige (2016) proposes a taxonomy based on noncompliant behaviors. He points out that noncompliance and “unobservability” are common features in all behaviors in unobserved economies. Accordingly, what distinguishes one unobserved economy from another is a particular rule being violated. Specifically, Feige offers the following definitions: “*unreported economy*” as when the noncompliance consists in the circumvention of the fiscal code by tax evasion; “*unrecorded economy*” as when the added informal value added arises from violations of the rules of national income accounting; “*illegal economy*” as the contravention of rules governing the production and distribution of prohibited goods and services (e.g., drugs, prostitution); “IE” as the circumventions of labor market regulations such as minimum wages, working conditions, social security, unemployment, and disability benefits.

Recent resolutions of the International Conference of Labour Statisticians (ICLS) of the ILO provide a valuable contribution to matching the differences among the concepts of the informal sector, employment, and economy. For instance, according to the ILO (2018b), the different scientific aims pursued by economists and statisticians are the source of the problem for a common definitional framework of the IE. Statisticians need a definition that makes it unproblematic to measure the phenomenon, on the other hand, economists look for a suitable definition for both analyzing causes and effects of informality and for providing effective policy implications. This heterogeneity of research purposes explains why, although the ILO (2015a) has proposed an “official” definition of IE in Recommendation No. 204 – “*all economic activities by workers and economic units that are – in law or in practice – not covered or insufficiently covered by formal arrangements*” – this concept is still not adequately followed by statistics. Indeed, the ILO publishes, almost exclusively, estimates on informal employment. As the ILO (2018b, p.22) explains, the reason is that “*the IE is not a statistical concept but a concept for policy purposes that embodies the sum of all parts of informality.*”⁴

While an exhaustive review of all the definitions of informality in literature is out of the scope of this article, in the following, I report the three most common sources of misinterpretations among “economic” and “statistical” definitions of the aggregates related to informality.

TABLE 1 Components of the informal sector, informal employment, and employment in the informal economy by institutional sectors in the Systems of National Accounts (SNA)

Enterprises/Economic units / Institutional Sectors	Institutional sectors		Jobs	
	Sub-sectors		formal	Informal
	General Government			
	Non-Financial Corporations			
	Financial Corporations		1	2
	Non-Profit Institutions serving Households			
	Households:	Formal	3	4
	Unincorporated enterprises	Informal sector	5	6
	Households:	Paid domestic workers	-	7
	Others	Production of goods for own final use	8	9
Employment in the informal sector = 5+6				
Informal employment = 2+4+6+7+9				
Employment in the informal economy = (5+6)+(2+4+7+9)				

Source: (Charmes, 2016, p.13).

The first source of confusion is due to the different meanings of the concepts of “IE” (mainly used by those who analyze informality as “informal value added”) and “informal sector” (typically used by scholars who examine informality in terms of “involved productive units”). The 15th ICLS resolution states that the informal sector consists of “units engaged in the production of goods or services with the primary objective of generating employment and incomes for the persons concerned. These units typically operate at a low level of organization, with little or no division between labor and capital as factors of production and on a small scale” (ILO, 1993, par. 5). This concept is dissimilar to the concept of informal employment which is, at least in developing countries, the most important source of the IE (as defined by ILO, 2015a). Specifically, informal employment and informal sector are conceptually different phenomena because, while informal employment refers to the characteristics of the job,⁵ the concept of informal sector refers to characteristics of production units. As a consequence, we can have formal employment in the informal sector (e.g., a formal job carried out in an informal enterprise) that should not be included in the IE, and informal employment in the formal sector (e.g., an employee of a registered enterprise whose employment relationship is not subject to labor legislation, income taxation, social protection, etc.), which should be included in the IE. Following Charmes (2016), Table 1 shows that: (1) the informal sector and informal employment are not mutually exclusive components of the labor force; (2) informal activities and jobs are positioned in the various institutional sectors of the Systems of National Accounts (SNA); (3) the informal sector and informal employment both contribute to employment in the IE.⁶

The second source of misunderstanding when comparing the labor-based ILO definition of IE to the definition of informality based on national account standards concerns the treatment of illegal activities. According to the 2015 ILO Recommendation No. 204, the IE does not cover illicit activities.⁷ On the contrary, the concept of NOE – introduced by the OECD (2002) and furtherly recalled by the 2008 SNA and European System of Accounts (ESA, 2010, par. 11.26)⁸ – includes (some) illicit activities. Specifically, the NOE includes both production of legal (i.e., underground and informal) and illegal activities, where the illegal production consists of those economic activities that fall within the national accounts production boundary, i.e., where both parties involved in an illegal economic transaction are willing partners, such as in prostitution, drug trafficking, and so forth.

The third source of misunderstanding in this literature concerns the treatment of tax evasion. In economic literature, the definition of IE (or shadow economy) often includes tax evasion. For instance, Schneider (2005, p. 600) defines shadow economy as including “*all market-based legal production of goods and services that are deliberately concealed from public authorities [...] to avoid payment of income, value added or other taxes, to avoid payment of social security contributions, to avoid having to meet certain legal labor market standards, such as minimum wages, maximum working hours, safety standards, and so forth, and to avoid complying with certain administrative procedures, such as completing statistical questionnaires or other administrative forms*”.⁹ This “economic” definition of informality – which focuses on the agent’s aim to avoid taxes and regulations – may diverge both from the ILO definitional approach – which focuses on the characteristics of economic units (as for the informal sector) or formal arrangements (e.g., the informal employment in the formal sector which should be included in the IE) – and also from the definition of informality based on national account standards – which focuses on the outcome of those activities (i.e., if they fall within the national accounts production boundary). On the latter discrepancy, it is worth pointing out that the difference between Schneider’s “economic” definition and Dell’Anno’s (2007) “NOE-based” definition of IE (see below in this section) depends on the different productive nature that the phenomenon of “tax noncompliance” (i.e., illegal and legal methods used by people or companies to reduce the tax they pay) may have. According to the SNA, tax evasion and tax avoidance are excluded from the NOE if they only generate a “tax-gap” (i.e., a difference between the total amount of tax theoretically collectible and the total amount of tax actually collected)¹⁰. On the contrary, if tax noncompliance behaviors consist of income or production underreporting, in this case, value added or production of these activities is included in the NOE as “*underground production for economic reasons*” because within the national accounts production boundary.¹¹

That of the ILO (2021) is one of the most recent attempts to find a harmonized single-definition of informality suitable for both “statistical” and “economic” purposes. It introduces two new concepts:¹² *IPA* (defined as “*all productive activities carried out by persons and economic units that are – in law or in practice – not covered by formal arrangements*” (ILO, 2021, p.7) and *informal market economy (IME)* (defined as “*all productive activities, carried out by workers and economic units for pay or profit that are – in law or in practice – not covered by formal arrangements*.” (ILO, 2021, p.11).

The concept of *IPA* has relevant advantages in terms of inclusiveness and flexibility with respect to the usual definitions of informal sector and informal employment. The key criterion to include productive activities within the aggregate of *IPA* is that they are not covered by formal arrangements. Accordingly, it would potentially include activities carried out by workers (independently of whether they are in an informal or formal job), (formal or informal) economic units, or by households, making this concept particularly suitable for measurement purposes. The second aggregate that the ILO (2021) suggests to include in the next ICLS resolution, i.e., the *IME*, consists of a subset of the *IPA*, as only activities taking place in relation to employment and enterprises producing for the market are included in the *IME*. According to the ILO (2021), this restricted definition allows us to outline a statistical aggregate that is more relevant for policy purposes (e.g., identify the size of the informal activities that can be formalized).¹³

Although ILO (2021) represents a valuable advancement in meeting policy and measurement needs, the treatment of unrecorded activities due to undeclared value added aimed to paying less taxes is still problematic. Explicitly, ILO (2021, Table 2) excludes under-declared value added due to tax elusion (i.e., legal methods to minimize the amount of tax) from *IME*, while ILO (2021,

TABLE 2 Matching different definitions of the informal economy

	Illicit production (Goods and services whose mere possession or consumption is subject to criminal law)	Under declared market value transactions of ordinary goods or services. (To pay less taxes i.e. tax elusion. This affects the amount of value added of market transactions)	Informal productive activities (All productive activities carried out by persons and economic units that are – in law or in practice – not covered by formal arrangements). It also includes non-declared transactions for part of the production declared in physical units, not just minimizing its value added (i.e. Tax evasion).
	Illegal economy		
Producer*			Informal Economy [all informal productive activities carried out by persons and economic units]
Type of Production *	Undeclared market production distinct from informal economy	For pay or profit	Not for pay or profit
Workers*	Not possible to become formal	Informal market Economy (All productive activities, carried out by workers and economic units for pay or profit that are – in law or in practice – not covered by formal arrangements) Liable to become formal or Already Formal (It includes Informal Employment - Employees are considered to have informal jobs if their employment relationship is, in law or in practice, not subject to national labor legislation, income taxation, social protection or entitlement to certain employment benefits such as advance notice of dismissal, severance pay, paid annual or sick leave (17th ICLS) - or Formal Jobs with partly informal activity.	Informal non-market economy Partial regulated or not at all but liable to be regulated (Own use, Unpaid trainee, Volunteer work)
Economic* Units	Unincorporated Criminal enterprises	Formal Sector (All economic units that are formally recognized as producers of goods and services and thus are covered by formal arrangements)	Informal Sector All producers of goods and services mainly intended for the market that are not formally recognized by government authorities as distinct market producers and thus not covered by formal arrangements. Household producing goods or services for own final use including direct volunteer work
ISTAT framework	Illegal Production (17) (Productive activities that generate goods and services forbidden by law or that are unlawful when carried out by unauthorized producers.)	Underground Production (T4+T5) (Productive activities that are legal but are deliberately concealed from public authorities to avoid payment of taxes, social security contributions, meeting legal standards or complying with administrative procedures)	Informal Sector production (T6) (Production undertaken by unincorporated enterprises in the household sector or other units that are unregistered and/or less than a specified size in terms of employment, and that have a market production)

Source: ILO (2021, Tables 1 and 2)

Source: OECD

^aSource: our elaboration based on: ILO(2021, Table 2).

^bThe issue of whether or not paid domestic workers should be included in the informal sector was left by the 15th ICLS as open for determination by the countries and, if included, it should be identified as a separate subcategory (OECD, 2002, p.168).

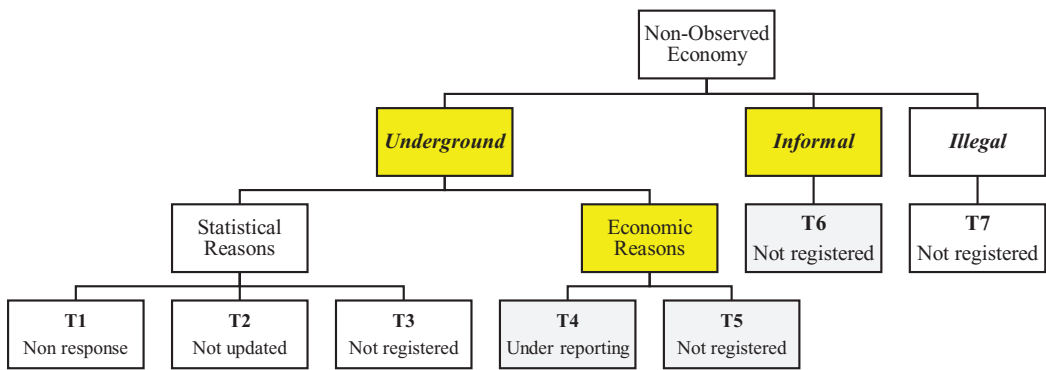


FIGURE 1 ISTAT framework of non-observed economy (NOE) [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/j.1368-1287.2021.16069.x)]

Table 1) includes tax evasion (i.e., illegal methods to pay less taxes) due to “*non-declared transactions for the portion of the production declared in physical units, not just minimizing its value added*”, within IME. This partition appears irrelevant for policy purposes and impracticable for statistical aims.

Chen and Carré (2020, p.5) point out that while the concept of informality has been criticized by many scholars “*as being too fuzzy and the formal-informal dichotomy as being too binary*,” it is unfeasible to renounce defining it. Indeed, it is unquestionable that workers and enterprises fall on a continuum between being fully formal or fully informal, therefore the formal-informal division is “*easy to criticize*.” But, as Harriss-White (2020) argues, a clear description of what should be and what should not be included within the IE remains pivotal for positive, normative, and measurement purposes. In line with Harriss-White’s (2020) argument, I consider Dell’Anno’s (2007) definition of IE as suitable for both statistical and economic purposes. By amending the Italian National Statistical Institute (ISTAT) analytical framework of NOE described in Calzaroni (2000) and OECD (2002), he delineates an aggregate of productive activities that may be suitable for economic analysis of informality. Specifically, along with the ISTAT framework, Dell’Anno (2007) defines the IE as a subset of the NOE that includes *underground production for economic reasons* (i.e., T4¹⁴ and T5¹⁵) and *informal production* (T6)¹⁶. In terms of productive activities involved in the IE, these can be (a) self-employment (including employers, own-account workers, and unpaid family workers) in informal (i.e., unregistered) enterprises; (b) formal (i.e., registered) enterprises that deliberately underreport their production to avoid the burden of regulation and taxation; (c) wage employment in informal jobs (i.e., without secure contracts, worker’s benefits or social protection). Figure 1 outlines the components of NOE according to the ISTAT framework and a subset of NOE (i.e., T4 + T5 + T6) that quantifies the informal value added suitable for an economic analysis of informality.

Following ILO (2021), Table 2 outlines the differences and overlappings between the most recent labor-based ILO (2021) definitions of informal aggregates (e.g., IPA, IME, IE, informal sector) and the concept of IE as derived by the ISTAT frameworks (i.e., informal production and underground production for economic reasons).

In conclusion, although in the last decade important steps forward have been made on the definitional issue, further interdisciplinary research is still needed to attain a unified definition of IE, suitable for both economic and statistical aims.

3 | THEORIES OF INFORMALITY

The IE is a complex and heterogeneous phenomenon, in which determinants and consequences depend on the characteristics of the involved units, and on macroeconomic and institutional factors. The multifaceted nature of IE explains the lack, in literature, of a comprehensive theory of informality suitable both for developing and developed economies. Indeed, a theoretical view that explains the emergence of informality as a strategy to survive in a developing country might be inapt to grasp the prevalent nature of IE in a developed country (e.g., Goel & Nelson, 2016). Because of that, to avoid misunderstandings among scholars, it is important to specify which type of informality a theory aims to explain.

Specifically, on the one hand, IE can be seen as a strategy of last resort to escape unemployment and poverty. The literature dealing with this type of informality mostly refers to it as an informal sector, rather than IE. It focuses on productive units (e.g., as a percentage of units involved) instead of the value added generated by these activities (e.g., as a percentage of official GDP). These informal productive units operate at a low level of organization and capital, they are geographically concentrated in low- and middle-income countries and, due to the low level of organization, this type of informality is strongly correlated with informal employment. According to this perspective, informality is a key topic both for labor and development economists and, given the focus on people, it also attracts anthropologists and sociologists generally interested in understanding the relationship between marginalization and informality (Wilson, 2010). In the wake of Hart (1973), the theoretical literature on this type of informality goes beyond economic analysis, because it involves research on human behavior, social exclusion, the role of social structures, and institutions.

On the other hand, informality may be approached by emphasizing its deviant behavior nature. According to this view, the IE (normally labeled in this literature as “shadow” or “undeclared” economy) is, first and foremost, considered a rational voluntary choice to “exit” from the formal economy to reduce taxation and regulation burden. This approach generally focuses on informality in terms of informal (or unobserved) value added (or GDP). According to this perspective, the analysis of informality has become a relevant topic in developed countries too. Indeed, in high-income economies, the analysis of the effects of the IE in terms of efficiency and equity is prevalent with respect to its consequences in terms of economic growth. From this viewpoint, research on informality mainly attracts public economists (e.g., to design effective public policies), national accounting institutions (e.g., aiming to adjust official statistics to achieve exhaustiveness of GDP), tax administrations, public finance researchers (e.g., to analyze tax evasion), and applied economists (e.g., dealing with the interactions between the IE and labor aggregate, firms’ competitiveness, etc.).

As for the definitional issue, where the difference between “statistical” and “economic” aims is a source of misinterpretation for the theoretical strand of this literature too, the heterogeneity of research goals (ranging from sociology to taxation theory) and the, at least, twofold nature of informality, have hindered the emergence of a consensus on a “general” theory of the IE.

In the following three sections, I systematize this literature according to three (partially overlapping) approaches and I direct my attention especially to the third one due to the lack of surveys in this strand.

Section 3.1 includes the literature that explores informality by analytical frameworks assuming the standard hypotheses of neoclassical economics (hereinafter “neoclassical approach”). In these theoretical models, productive units are rational agents that voluntarily choose to

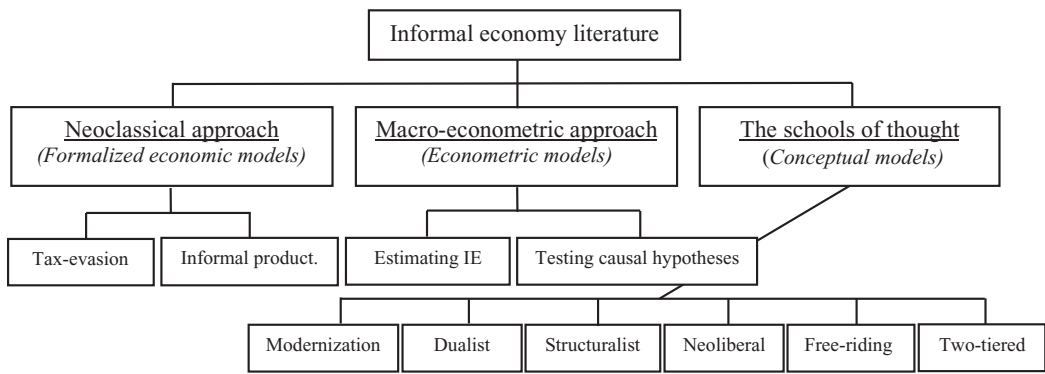


FIGURE 2 A roadmap of the literature on informality

participate in the IE by maximizing a welfare (e.g., utility, profit) function subject to resource constraints. The neoclassical approach includes both microeconomic and macroeconomic formalized theories where, in addition to the assumption of rational choice, the steady-state equilibrium that defines the aggregate size of the IE depends on the allocation of resources among formal and informal units in a competitive market.

Section 3.2 includes those empirical studies dealing with both measurement issues and testing causal hypotheses on the determinants of the IE. These analyses are predominantly explored by macro-econometric exercises (hereafter “macro-econometric” approach) and explore the phenomenon of informality mainly in terms of value added as a percentage of official GDP.

Section 3.3 surveys the studies that systematize the theoretical approaches to the IE (hereinafter “schools of thought”). This literature has a multidisciplinary character and pays particular attention to the relationship between informal and formal sectors.

Figure 2 summarizes the proposed structure of the literature survey.

3.1 | The neoclassical approach to the theory of the IE

Under the neoclassical approach to the IE, I include those analytical microeconomic models in which the rational agent chooses to operate informally as a result of a constrained maximization of his/her value function and, in macroeconomic models, the equilibrium is obtained by assuming a competitive market. These theoretical frameworks usually follow two main analytical workhorses, depending on the predominant nature of the IE under scrutiny. If the analysis examines the IE “underground” component (i.e., underreported earnings deliberately concealed to avoid the burden of taxation), then the mainstream framework is Allingham and Sandmo’s (1972) model of tax evasion (e.g., see for a survey Alm, 2019; Devos, 2014). Conversely, if the focus is on the “informal sector,” where informality is mainly related to employment, then the usual analytical framework consists of macroeconomic two-sector search and matching models. Among several studies proposing macroeconomic frameworks where the size of informality depends on the allocation of the labor force between formal and informal sectors, Rauch (1991) plays a significant role. He develops a model where a firm’s decision to hire informal workers depends on the desire to remain small to enjoy legal exemption from a mandated minimum wage policy. Ihrig and Moe (2004) analyze the agent’s decision to accumulate capital and to work in the formal and informal sectors using a two-sector dynamic equilibrium model. In terms of policy implications,

the authors conclude that to reduce the informal sector the most effective policies are to cut tax rates and increase enforcement by implementing higher tax penalties for informal agents, rather than increasing the chances that an informal agent may be caught. Amaral and Quintin (2006) propose a dynamic model where, in equilibrium, the characteristics of formal and informal workers systematically differ, even though labor markets are perfectly competitive. This implies that the hypothesis of formal barriers to movement between sectors is not important to explain the emergence of informality. According to these results, Amaral and Quintin (2006, p.1552) conclude that the “*dualistic view of labor markets in developing countries should be questioned.*” Similarly, Allen et al. (2018) propose a multi-industry general equilibrium model where formal and informal firms often operate in the same markets, and the degree to which informal firms are similar to formal firms in terms of productivity is heterogeneous across industries. In this framework, informality tends to decrease with industry productivity and establishment size. Günther and Launov (2012) empirically test whether employment in the informal sector in a developing country (i.e., Côte d'Ivoire) is a workers' voluntary choice based on utility maximization or rather a strategy of last resort to escape unemployment. This query indirectly deals with a fundamental question in this literature, whether “*individuals are poor because they are employed in the informal sector (an implication of a segmented labor market); or alternatively, whether they are employed in the informal sector because they are poor(ly endowed) with characteristics that generate high returns in the formal sector (an implication of a competitive labor market)*” (p.88). Günther and Launov (2012) finds that both voluntary and involuntary employment describes the empirical data. Dell'Anno (2018) proposes a microfounded macroeconomic modeling framework, in which, other than taxes, the worker's choice to informally operate depends on indivisible entry costs, credit market imperfections, and, the composition of the labor force (i.e., the proportion of workers who do not have the opportunity to work in the IE, such as employees). Ulyssea (2018) develops a competitive equilibrium model where heterogeneous firms can exploit two margins of informality: an “extensive margin” (when firms do not register their business); and an “intensive margin” when firms pay workers “off the books.” According to this framework, two key factors explain the firm's decisions regarding entry, production, and compliance: the firm's heterogeneity in terms of productivity and the entry costs into the formal sector. The main result of this research is that within the same economy three different types of informal firms co-exist (“survival firms,” which are too unproductive to operate formally, even if entry costs were removed; “parasite firms,” which are productive enough to survive as formal firms once entry barriers are removed but choose to remain informal; informal firms (“*de Soto's view*”), which are kept out of formality by high entry costs, but if these were removed, they would become formal and improve their performance). In conclusion, I suggest Elgin and Ertuk (2019) for a recent survey of analytical models on informality.

3.2 | The macro-econometric approach to the IE

The second strand of this literature analyzes the IE by following a macro-econometric perspective. Since their early papers, Cagan (1958), Gutmann (1977), Feige (1979), Tanzi (1980), and Gaertner and Wenig (1985) have paid particular attention to the issues of measurement and analysis of the IE determinants. These studies normally focus on the value added concealed from official statistics rather than the number of people involved in informal activities. Accordingly, this perspective tends to overweight the underground production (i.e., undeclared GDP due to economic reasons) with respect to the informal production in the analysis of the IE. From a theoretical background,

these empirical studies encompass theoretical hypotheses from economic models both on tax evasion and on labor or development literature.

The macro-econometric approach has probably been the most fast-growing strand in the literature on IE in the last 20 years (e.g., Afonso et al., 2020; Dell'Anno, 2007; Elgin & Ertuk, 2019; Feld & Schneider, 2010; Görxhani, 2004a; Goel & Nelson, 2016; Jessen & Kluve, 2021; La Porta & Shleifer, 2014; Loayza, 2018; Medina & Schneider, 2018; Ohnsorge & Yu, 2021; Schneider, 2011; Schneider & Buehn, 2017; Schneider & Enste, 2000; Schneider & Williams, 2013; Ulyssea, 2020). One of the reasons for this success is the relatively recent abundance of estimates of the size of informal GDP. These estimates have encouraged researchers to empirically explore both the most significant causes of informality and also the effects that IE has on other variables.

Taking into account that an exhaustive list of these studies is outside the scope of the present survey, Dell'Anno (2021a) groups potential drivers of the IE into six categories. These categories deal with: taxation system (e.g., measured as overall tax burden, tax mix, tax complexity, marginal tax rate); regulatory system (e.g., overall regulatory burden, employment protection legislation); labor force composition (e.g., self-employment rate, unemployment rate, the existence of a significant amount of illegal immigrants, unskilled workers); enforcement system (e.g., probability of detection, fine rates, the strength of criminal penalties); tax morale and institutions (e.g., historical roots, fairness of public policy, corruption, quality of political institutions, media and economic freedom).¹⁷

An important issue for these empirical analyses is to determine the direction of causality between IE and the macro-variables. Without any claim to exhaustiveness, among the studies that focus on the relationship between IE and other economic aggregates, I quote Ferreira-Tiryaki (2008), Granda-Carvajal (2010), and Elgin et al. (2021) on the relationship between informality and business cycle; Afonso et al. (2020) on the link with economic growth; Bonnet and Venkatesh (2016) on poverty; Chong and Gradstein (2007) and Dell'Anno (2021b) on inequality; Capasso & Jappelli (2013) on financial development; Dreher and Schneider (2010) and Goel and Saunoris (2014) on the relationship between IE and corruption.

In conclusion, although several studies agree on both the identification of potential drivers of IE (e.g., tax policies and state regulation according to Schneider, 2019) and on the idea that the size of their marginal effects depends on the economic development of the country (e.g., Goel & Nelson, 2016), the doubts on both the reliability of the estimates of the IE and the issue of endogeneity leave an aura of suspicion on the consistency of these empirical findings.

3.3 | The Conceptual approaches to (or “schools of thought” on) the IE

In the third strand of literature, I include those researches that, following the seminal studies of ILO (1972) and Hart (1973), aim to explore the emergence and evolution of the IE and its relationship with the formal economy. In this strand, the concept of informality is mainly focused on the notion of the informal sector (i.e., “units typically operate at a low level of organization, with little or no division between labor and capital as factors of production and on a small scale”, ILO, 1993, par. 5) and, consequently, the proposed conceptual views are more suitable to explaining informal employment than unreported value added.

The contribution of this section to the literature is twofold. First, I survey the main studies that deal with the classification of the theoretical views on IE (e.g., Chen, 2012; Gerry, 1987;¹⁸ Chen et al., 2004; Hart, 1973; ILO, 2015b; La Porta & Schleifer, 2008; Perry et al., 2007; Tokman, 1978; Williams, 2008, 2015; Williams & Kedir, 2018; Williams & Nadin, 2011), by following a

chronological order. Second, given that these studies use different labels, hypotheses, and references, at the cost of smoothing some minor distinctions, I combine the existing theories into six “schools of thought” in Section 3.3.1. These six conceptual approaches are defined by aggregating the (not necessarily mutually exclusive) theoretical views of informality according to a criterion of homogeneity in terms of causes, characteristics of informal units, and policy implications.

Bangasser (2000) considers Hart (1973) and ILO (1972) as milestones in this field of research because they propose a new viewpoint for the informal sector that disagreed with the prevalent “miserabilist” views of survivalist activities of working poor in urban areas. ILO (1972, p.5) highlighted as “*far from being only marginally productive*” – as supposed by the old-fashioned concept of the “traditional sector” – [the informal sector] *is economically efficient and profit-making, though small in scale and limited by simple technologies, little capital, and lack of links with the other (“formal”) sector.*” Moreover, the employment in the Kenyan informal sector “*has probably increased a good deal faster in the informal than in the formal sector*” (ILO, 1972, p.6). This evidence conflicted with the first generation of development economists¹⁹ (also known as the “Modernization” view) that predicted that the development of the (formal) market economy would lead developing economies on the same successful pathway followed by Western capitalist countries after the Second World War. Unfortunately, for developing countries, the economic growth was lower than expected and, since the 1970s, it became clear as the theoretical prediction of a declining informal sector was destined to remain a mirage. Historically, by rejecting the hypothesis of informality as a transitory phenomenon that would disappear with economic growth, these studies have opened the scientific debate on the theories of the IE.²⁰

Following a chronological order, Tokman (1978) is among the first authors to review the theories of informality by focusing on the relationship between the informal and formal sectors. He distinguished two main theoretical approaches to informality (i.e., a “*benign*” and a “*subordinate*” view) and proposed a third integrated approach, which he labeled as “*Heterogeneous subordination*” of informal on the formal sector. Along with the first approach, the relationship between informal and formal sectors is “*benign*” because the informal sector has “*the potential for a more adequate pattern of growth with a more egalitarian distribution of income*” (Tokman, 1978, p.1066). Tokman (1978) indicated two leading hypotheses to explain the economic rationales behind the “*benign*” relationship. According to the “*duality sub-approach*,”²¹ the informal sector is an autonomous self-contained segment (i.e., without significant links to the formal sector) characterized by providing employment, goods, and services for the lower-income groups of the urban population. It is economically efficient and has comparative advantages to similar activities operating in the formal economy. The second hypothesis supporting the *benign* approach sees the informal and formal sectors as complementary. Consistent with the “*complementary sub-approach*”²², the informal sector “*is highly integrated to the rest of the economy exporting three quarters of its production and importing a similar proportion of its consumption*” (Tokman, 1978, p.1067). Such integration is “*benign*” since the informal sector predominantly produces distinct and specialized goods and services, that is, labor-intensive productions that employ low-skilled workers. For this reason, under the complementarity approach, the size of the informal sector is expected to expand with the size of the formal sector and the presence of informal activities does not generate negative externalities as a result of unfair market competition between the two sectors.

A second conceptualization of the relationship between the formal and informal sector is in opposition to the “*benign*” relationship approach. Tokman (1978) defined this view as a “*Subordination*” approach. He outlined two sub-approaches to explain how a subordination of the informal sector to the formal sector is detrimental for economic development. The “*marginalization*” hypothesis assumes that the informal sector operates mainly through the lack of access

to resources of production and product markets (e.g., Quijano, 1974). As Tokman (1978, p.1069) argued, *“the lack of stable access to basic resources of production, because they are monopolized by the formal sector, determines that informal activities can only operate around residual resources, being excluded from the possibility of technological improvement.”* The second sub-approach assumes an *“exploitative integration”* of the informal sector into the capitalist system (e.g., Gerry, 1974). This view posits a scenario inspired by Marx's theory on the role of the *“reserve army of labor”* in capital accumulation. In this perspective, the informal sector may be seen as the pre-capitalistic sector and the formal economy as the capitalistic sector. Consistent with the Marxian model, informal workers constitute the *“reserve army”* that facilitating capital accumulation by releasing cheap labor to the formal capitalist is functional to the capitalistic system. As Mkandawire (1985) explains, the informal sector facilitates the exploitation of the working class to weaken the bargaining position of formal workers thus forcing their wages down. Tokman pointed out an additional cause through which the informal sector favors capitalist accumulation compared to those originally assigned by the Marxian model to the *army of labor reserve*. Indeed, the informal sector also subsidizes the consumption of urban workers by providing survivalist activities. Thereby the capitalistic (formal) sector can pay lower real wages and lower taxes to sustain and reproduce the proletariat and, consequently, capitalists earn higher profits.

Tokman (1978) concluded his analysis by proposing a framework that is not entirely different from the two previous approaches but has common features with both of them. Tokman (1978) labeled it as *“heterogeneous subordination”* because it is much closer to the *“subordination”* than to the *“benign”* relationship approach. Specifically, it *“implies a subordinated relationship for the sector as a whole, but resulting from different processes occurring within it”* (Tokman, 1978, p.1065). This conceptual framework considers the *“informal sector neither as a completely integrated nor as an autonomous sector, but rather as one with significant links with the rest of the economy, while simultaneously it also presents a considerable degree of self-containment”* (Tokman, 1978, p.1071).

Chen et al. (2004) outline one of the most known classifications of the theoretical approaches to IE. In this study, the authors delineate three fundamental schools: the dualist, the structuralist, the legalist. Chen (2012) enriches this tripartition by two additional views (i.e., the voluntarist and the holistic models). Consequently, according to Chen's (2012) taxonomy, we distinguish five theoretical views.

The *Dualist school*, which widespread in the 1970s mainly through ILO (1972) and Hart (1973). It considers the informal sector as an autonomous segment of the economy characterized by providing employment, goods, and services for the lower-income groups of the urban population. This school corresponds to Tokman's (1978) *“duality sub-approach”* under the *“benign”* view. Accordingly, informal activities are autonomous, economically efficient and have comparative advantages to similar activities developed in the formal sector. According to the *Dualists*, the links between informal enterprises and government regulations have minor relevance because informal and formal activities operate *“as a distinct separate sector of the economy”* (Chen, 2012, p.5). The macroeconomic causal root of the informal sector depends on the difference between the growth rate of labor demand for modern (formal) jobs and the urban workforce's faster growth rate – enhanced in developing countries by the growing population and urban migration. This mismatch in the labor market is intensified by the discrepancy between the excess of low-skilled urban unemployed and the demand for modern job opportunities.

The *structuralist school* was popularized by Moser (1978) and Castells and Portes (1989). According to this approach, informal activities are subordinated economic units (micro firms) and workers that serve to reduce input and labor costs and, by doing so, they increase the competitiveness

of large capitalist firms (Chen et al., 2004). For the structuralists, the informal sector is inextricably connected and interdependent to formal economy. This school shares several aspects with the “*exploitative integration*” within the “*subordination approach*” proposed by Tokman (1978). Indeed, both the views see the informal sector as highly integrated into the formal economy and, unlike the dualists, they argue that the persistence and the growth of the informal sector depend on “*the nature of capitalist development (rather than a lack of growth)*” (Chen et al., 2004, p.17).

The *Legalist school*, inspired by de Soto's (1989, 2000) analysis of Latin American countries, was very influential in the economic debate on informality in the 1980s and 1990s. Consistent with this view, informal activities are “*‘plucky’ micro-entrepreneurs who choose to operate informally in order to avoid the costs, time, and effort of formal registration*” (Chen, 2012, p.5). Unlike the dualist and the structuralist schools, the legalists argue that informal activities were neither a function of poverty nor a structural necessity of late capitalism. Instead, informality is a choice made by rational actors facing a legal environment that hampers business activity (Bonnet & Venkatesh, 2016). These activities circumvent the legal system because the formal institutional framework, taxation, and administrative costs of the formal economy are counterproductive to the growth of small and medium-sized enterprises. According to de Soto's perspective, informal entrepreneurs represent efficient market forces that emerge as a reaction to over-regulation and government oppression (Biles, 2009). As stated by Chen (2012, p.5–6), in terms of policy implications, the legalist school suggests simplifying “*bureaucratic procedures to encourage informal enterprises to register and extend legal property rights for the assets held by informal operators in order to unleash their productive potential and convert their assets into real capital.*”

Chen (2012) defines a fourth approach to the informal sector labeled as *Voluntarist school*. It is based on Maloney's (2004) analysis of informality in Latin American countries and focuses on the intentional and rational choice of informal entrepreneurs to avoid regulations and taxation. Unlike the “*legalists*,” the *voluntarists* do not consider the size of informality as predominantly due to cumbersome registration procedures and as a “*plucky*” business activity. On the contrary, this school argues that informal enterprises create unfair competition for more efficient formal enterprises because informal firms can gain a price advantage by not paying taxes and not complying with regulations. For Maloney (2004, p.1173) “*the urban informal microenterprise should be viewed as a part of a voluntary small firm sector similar to those in advanced countries that, due to the laxity of enforcement of labor and other codes, is able to choose the optimal degree of participation in formal institutions*”. In terms of normative implications, Chen (2012, p.6) highlights that for the voluntarists: “*informal enterprises should be brought under the formal regulatory environment in order to increase the tax base and reduce the unfair competition to formal businesses.*”

Chen (2012) proposes a fifth category of theoretical approaches to informality: the *Holistic Conceptual Models*. These models, which focus on the heterogeneous composition of the informal sector, consider various factors driving informality. Among these integrated approaches, the author cites the “*multi-segmented model of informal employment*” developed by the Women in Informal Employment: Globalizing and Organizing (WIEGO). The WIEGO approach renounces the ambitious goal to outline a general theory of informality to define informality in terms of employment status. In particular, this definition, based on two key dimensions, that is, the economic risk (of losing a job) and the authority (over the establishment and other workers), is functional for policy purposes (Chen, 2012). A second *holistic* model of informality is based on Perry et al. (2007). In line with this view, informality may have different forms and causes. It may be seen as an “*Exit*” option for formal activities (that is, a voluntary choice to work in the IE)²³ and as a result of an “*Exclusion*” from the formal sector, (i.e., the poorest and most marginalized people who do not

have access to the formal sector need to work informally as a survival strategy for them and their families).

La Porta and Shleifer (2008) belong to a growing literature (e.g., Williams and his co-authors, Ulyssea, 2018, 2020) that aims to determine which theoretical approaches to informality are empirically supported. In particular, La Porta and Shleifer (2008) consider three alternative views to classify informal activities (specifically, they refer to informal firms) based on characteristics of productive units and test, which of these views better explains informality in developing countries.

The *Romantic view* assumes that informal firms are “extremely productive but are held back by government taxes and regulations, as well as by lack of secure property rights and access to finance.” (La Porta & Shleifer, 2008, p.276). The key hypothesis of the “Romantics” is inspired by de Soto (1989) because informal firms are considered similar in terms of efficiency to formal ones but are kept down by invasive public policy.

The other two views are more skeptical about the potential efficiency of informal firms because they are run by entrepreneurs with lower human capital.

The *Parasite view* considers IE as a collection of firms that need to stay small to avoid detection and therefore they lack the necessary scale to produce efficiently (Baily et al., 2005; Farrel, 2004). According to Farrel (2004, p.28) “*the cost advantage that informal companies gain by avoiding taxes and regulations more than offsets their low productivity and small scale.*” Thus, informal firm “*hurts growth both because their small scale makes them unproductive and because they take away market share from bigger, more productive formal competitors*” (La Porta & Shleifer, 2008, p.277). Consistent with the *Parasite view*, the voluntary “exit” choice to operate informally is a source of a negative externality, therefore it should be tackled through better enforcement by the policymaker.

The *Dual view (or Wal-Mart theory)* is intimately related to the “traditional” theories of development economics, which see IE as a transitory phase of the transition, from the rural, preindustrial to the urban and industrial (formal) economy (e.g., Lewis, 1954; Rosenstein-Rodan, 1943; Rostow, 1960; Murphy et al., 1989). La Porta and Shleifer (2008) highlight the differences between the *Wal-Mart* theory and the previous approaches: “*unlike the romantic view, the dual view predicts that informal activities are very different from the formal ones in their characteristics not affected by government policies (p.277) [...] Unlike the parasite view, the dual view does not see the unofficial firms as threatening the official ones, because they are hugely inefficient and hence unlikely to be able to charge lower prices for the same products (p.278).*”

La Porta and Shleifer (2008) test which of these views better explains the informality in developing countries using data from the World Bank surveys of individual firms. According to their evidence, informal firms are small, use less capital, and are extremely unproductive, compared even to the small formal firms. This result is consistent with the *Wal-Mart* view. In terms of policy implications, the authors conclude that “*one should not expect much from the unofficial economy, with its millions of entrepreneurs, except to hope that it disappears over time*” (La Porta & Shleifer, 2008, p.347). In other words, by revitalizing the basic idea of “Modernization” theories, they sustain that the reduction of informality should only rely on “*the hope of economic development lies in the creation of large registered firms, run by educated managers and utilizing modern practices, including modern technology, marketing, and finance.*” (La Porta & Shleifer, 2008, p.279–280).

Ulyssea (2018) provides a unified theoretical framework to show, as La Porta and Shleifer’s (2008) views are not alternative perspectives but coexist in IE because they reflect the heterogeneity in the firms’ productivity. Using the Brazilian IE as a case study, he finds that the *Romantic* view explains 16.8% of all informal firms, *Parasite* firms correspond to 38.7%, and the remaining informal firms, too unproductive to ever become formal and are only able to survive in the informal

sector – that is, *Dual economy* view, represent the majority of informal firms (44.5%). Although Ulyssea's empirical evidence is consistent with La Porta and Shleifer's (2008) result that informal firms are extremely unproductive, compared even to the small formal firms, the studies diverge from a normative viewpoint. For La Porta and Shleifer (2008, p.279), the differences in productivity between formal and informal firms are so wide “*that it is hard to believe that simply registering unregistered firms would eliminate the gap*” as well as “*...stronger enforcement would devastate the livelihood of millions of people surviving near subsistence.*” According to the authors, informal firms will die as the economy develops and “*the formal firms created will be new firms run by new people, not previously informal firms*” (p.278). On the contrary, Ulyssea (2018, p.2045) considers a transition to formality as possible, indeed “*informality reductions can be but are not necessarily associated to higher GDP, TFP, or welfare.*”

Williams and his co-authors have proposed three comparable classifications of the conceptual approaches to informality. Williams (2008) advised a categorization that, in the wake of Tokman (1978) and Chen et al. (2004), focused on the relationship between informal²⁴ and formal economy.²⁵ Williams and Nadin (2011), Williams (2015), and Williams and Kedir (2018) reformulated Williams's (2008) classification by using different labels for (seemingly) similar theoretical approaches. In the following, I match these taxonomies for the sake of brevity, in order to outline four conceptual approaches.

The *Modernization theory* takes its name from traditional theories of economic development (e.g., Geertz, 1963; Lewis, 1954; Rostow, 1960). In this view, the IE is a product of economic underdevelopment and, due to its transient nature, it is destined to disappear with economic growth. The *Modernization theory* corresponds to the category of informality as “*a residue*” in Williams (2008) and, in general, it is inspired by Tokman's (1978) “*duality sub-approach*.”

The *Structuralist theory*²⁶ (e.g., Castells & Portes, 1989; Davis, 2006; Portes, 1994) considers IE as a “*direct by-product of the advent of a deregulated open world economy*” (Williams, 2015, p.336). Williams and Kedir (2018, p.158) point out as informal firms are “*the outcome of too little state intervention in not only the economy but also social protection and social transfer systems.*” Consistent with this theory, the global economic system exploits informal workers, who are excluded from formal jobs and welfare support and work informally as a last resort. In terms of policy implications, Williams and Kedir (2018), following Davis (2006), state that, for the structuralists, the lack of state intervention to protect workers and citizens may explain cross-country variations in informality.

The *Neoliberal theory*²⁷ (inspired by de Soto, 1989, 2000) hypothesizes that the IE is “a populist reaction to high taxes, a corrupt state system and too much interference in the free market, resulting in workers making a rational economic decision to voluntarily exit the formal economy in order to avoid the costs, time, and effort of formal registration” (Williams, 2015, p.336). According to this perspective, the policies to reduce informality are in contrast with the structuralists' recommendations. Specifically, for the supporters of the “neoliberal theory,” the best policies to reduce informality consist in lowering taxes, promoting deregulation, and minimal state intervention.

The *Post-structuralist approach* includes different hypotheses that share a common critical position on both the structuralist and neoliberal explanations. The post-structuralists argue that IE is not (only) a rational choice of economic actors influenced by the cost/benefit ratios (i.e., neoliberal theory) or an exploitative global economic system (i.e., structuralist theory), rather, it is a voluntary choice made by marginalized people. This “exit” option can be due to social relations, redistributive reasons, or resistance practice to the exploitation of workers as well as identity reasons.²⁸ Among the post-structuralists, Biles (2009, p.232), includes Whitson (2007) who shows as “*informality not only served as a simple livelihood strategy; many men and women employed informality*

as a form of resistance in response to the exploitation of workers within Argentina's neoliberalized economic system".

Williams and Kedir (2018) propose an *Institutional approach*, which replaces Williams and Nadin's (2011) *Post-structuralist* approach. As the *Post-structuralist* view, the *Institutional approach*, despite including heterogeneous hypotheses on the causes of informality, shares the disagreement with the neoliberal and structuralist views. For the *institutionalists*, IE is a consequence of the discrepancy between formal and informal institutions (e.g., Feige, 1997; Godfrey, 2015; Webb et al., 2014; Williams et al., 2015). The basic idea of the *institutionalists* is inspired by Feige's (1997, p.22) conclusion that "when formal and informal institutions clash, noncompliant behaviors proliferate, forming various underground economies." As Williams and Kedir (2018) point out, informal activities sit outside the formal framework, because they distrust formal institutions, but within the norms, values, and beliefs of informal institutions. Consequently, "the greater the incongruence between formal and informal institutions, the greater is the likelihood of businesses being unregistered" (Williams & Kedir, 2018, p.158).

Williams and Kedir (2018) test the relevance of these approaches to explain cross-country variations in the level of nonregistration at start-up in 142 countries. They conclude that *Modernization*, *Political economy*, and *Institutional* approaches may explain differences in informality but not the *Neoliberal theory*.

Another strand of literature classifies the informality conceptual frameworks by focusing on the drivers of the decision to participate in the IE (e.g., Perry et al., 2007). One of the most influential documents in terms of normative implications, ILO's (2015b) guide on national employment policies, follows this approach. It provides the theoretical ground for policy recommendations and describes four main theoretical approaches to informality.

The first view sees informality as "*exclusion*." According to this perspective "informal workers and firms would like to operate in the formal economy, they do not have the opportunity to do so and stay in the IE to make a living" (ILO, 2015b, p.7). For the supporters of the "*exclusion*" view, IE originates from a lack of demand in the formal labor markets and it increases with poverty, lack of education, and gender-based constraints.

As opposed to the *exclusion* view, informality may be seen as a voluntary choice to "*exit*" from the legal framework. The *exit* option is based on a rational choice: workers weigh the benefits of working formally against those of working informally, selecting the form of employment with the greatest net benefits (e.g., Maloney, 2004). Following Loayza and Rigolini (2011), the choice to operate in the IE is determined by the opportunity costs of formality (in terms of regulatory and tax burden, quality of public services) and the distribution of the workers' skills. From a normative viewpoint, under the theoretical umbrella of the "*exit*" option, there are both scholars who see informal firms as "*parasitic*"²⁹ – calling for strict enforcement of regulation – and researchers who consider IE "*as a way to avoid high entry costs, suggesting that strict enforcement may hold back a dynamic part of the economy*"³⁰ (ILO, 2015b, p.7).

A third view sees informality as "*driven by market competition*." Consistently with this perspective, the main determinant of informality is the need to lower labor costs and protect profitability in a competitive global economy as supposed by Castells and Portes (1989). Unlike the "*exclusion*" standpoint, the *market view* hypothesizes that IE is due to an existent demand for informal labor rather than a lack of demand in the formal labor market. The key hypothesis of this approach largely agrees with the *Structuralist* view (Chen, 2012) or the *Political economy* theory (Williams, 2015).

ILO (2015b) defines a fourth view of the IE, the "*Two-tiered*" approach to informality (Fields, 1990; Perry et al., 2007). In line with Chen's (2012) *Holistic approach* or Williams and Nadin's

(2011) *Post-structuralist* approach, this perspective is an attempt to gather the “*exclusion*” and voluntary “*exit*” views into a single approach. Fields (1990) underlined as two qualitatively distinctive branches co-exist in the informal labor market economy: a lower-tier, characterized by easy entry and low-wage unorganized and unprotected work (i.e., the marginal activities due to “*exclusion*”); and an upper-tier of the labor market, with restricted entry to self-employment activities, where we find people who voluntarily choose to exit from the formal economy. These two segments of the labor market differ in terms of rationales. On the one hand, people in lower-tier activities seek to get out of them, aspiring to engage in the upper-tier segment of the self-employed. On the other hand, the upper-tier segment of the IE is largely related to tax and regulatory evasion and, as Field (1990, p.65) remarks, “*given the constrained choices available to them, a great many of informal sector workers are in that sector voluntarily.*”

Several empirical analyses test the prevalence of “*exclusion*” or “*exit*” choice in informality across countries (e.g., Gërxfhani, 2004b; Maloney, 2004; Oviedo et al., 2009). They find that the “*exclusion*” hypothesis is more relevant in developing countries while the “*exit*” in developed economies. Perry et al. (2007) affirm that the “*exclusion*” and “*exit*” hypotheses coexist in the economy, but the former is more applicable to waged informal employment, while the “*exit*” option more accurately characterizes own-account informal workers (i.e., self-employed with no employees). Chen (2012, p.11) concludes her overview of the theoretical approaches to IE, pointing out that “*most causal theories are valid – but only for certain segments of informal employment; and no single causal theory can explain each segment of informal employment.*” Similarly, Williams (2008) concluded his review of the literature with two main deductions: “*there is no one universal relationship between formal and informal work but instead, many different relationships existing in various parts of the picture*” (p.117); “*we cannot simplistically attribute one form of work with positive attributes [i.e., formal] and the other [i.e., informal] with negative attributes, and nor can one treat them as discrete and separate forms of work.*” (p.119). Williams et al. (2015, p.297), rather than describing informal workers “*as either universally exit- or exclusion-driven, as is the case in these two conventional approaches, a second wave of thought has adopted a more nuanced understanding. This evaluates the ratio of exit-driven to exclusion-driven informal workers.*”

To conclude this review, at the state of art, there is not a universally applicable theory of the IE. We have several, but not mutually exclusive explanations with diverse causal hypotheses. In terms of normative implications, it implies that effective policies have to be tailored to the country-specific composition and nature of the IE.

3.3.1 | A conceptual framework of the “schools of thought” on the IE

This section aims to contribute to the literature by outlining a conceptual framework able to encapsulate into six approaches the main features of the schools of thought on the IE exposed in Section 3.3.³¹ Tables 3 and 4 reports the differences among these approaches according to eight dimensions: the key idea of the theory (“*General view*”); the main characteristics of informal activities (“*Types of activities*”); the principal rationales to operate informally (“*Causal roots*”) and the forced or voluntary nature of choice (“*Nature of choice*”); the “*Policy implications*”; and what the theory hypothesizes (if it does) on the relationship between formal and informal economies (“*link to the formal economy*”)³²; in what type of countries the approach is mostly used (“*Prevalent in*”); the authors who have developed the theoretical approach (“*Proposer(s) of the theory*”).

In our view, although this synopsis omits some details of the original classifications, it provides a unified conceptual framework useful to move among different scholars’ classifications. The last

TABLE 3 Conceptual framework of the “Modernization”, “Dualist” and “Structuralist” approaches on the informal economy (IE)

	Modernization	Dualist	Structuralist
General view	The informal sector is seen as a stage of the evolutionary process of the market economy. The informal activities will disappear as a result of economic progress in developing countries.	The informal sector is a premodern, distinct and persistent sector acting as an intermediate space between the mainstream (modern) formal system and complete unemployment.	The IE is a structural feature of modern capitalist development. Capitalist firms attempting, acting with the complicity of the state, to reduce wages, labor safety standards and enhance flexibility by exploiting unprotected informal workers.
Types of activities	Workers with lower human capital work in informal and smaller firms and receive lower wages, whereas those with higher human capital are allocated to the larger and more productive firms and receive higher wages.	Survivalist activities by the working poor with few (if any) links with the formal economy.	Vulnerable workers, such as women, youth, immigrants, and unskilled are exploited by formal firms.
Causal roots	The informal sector is due to the excess of labor supply (unskilled rural migrants). These “excluded” informal workers will find a job in the formal economy with the modernization of the economy.	Informal workers are excluded from modern economic opportunities. The informality is due to the mismatch between people’s skills and the structure of modern economic opportunities (Chen, 2012). Especially in developing countries, informality is enlarged by imbalances between the growth rates of the urban population and formal labor demand.	The IE is a side-effect of capital accumulation and globalization. It is not a stage in the transition from a traditional to modern society. It does not depend on the inability of the formal economy to absorb workers, or on some voluntary decision of self-employed.
Nature of choice	Exclusion-driven	Exclusion-driven	Exclusion-driven
Policy implications	Depending on the specific theories of economic development. In general, it is sufficient to develop the formal economy (e.g., by a “Big push,” higher capital accumulation and saving rates, more international trade, etc.)	More job opportunities, better credit and business development services to informal operators, enhanced infrastructure and social services to families of informal workers.	More regulation of commercial and employment relationships between the informal and formal economies in order to address unequal relationships between “big business” and subordinate producers.

(Continues)

TABLE 3 (Continued)

	Modernization	Dualist	Structuralist
Link to the formal economy	Counter-cyclical relationship. The informal sector declines with the growth of the formal sector. Informal workers will be absorbed by the formal economy with economic development.	Undetermined. According to Hart (1973: 87) “the possibility that the informal economy [...] may be developing at a rate faster than other sectors of the national economy. [...] cannot be dismissed on a priori ground.”	Pro-cyclical relationship. Informal and formal modes of production are inextricably linked within the same economic system (Chen et al., 2004).
Prevalent in	Developing countries	Developing countries	Developed countries
Proposer(s) of the theory	Lewis (1954), Rostow (1960), Harris and Todaro (1970), Rauch (1991).	Hart (1973), ILO (1972).	Moser (1978), Castells and Portes (1989), Portes (1994), Thomas (1995).
Different labels used for comparable theoretical view	Subordination due to marginalization (Tokman, 1978), dual view (La Porta & Schleifer, 2008), residual (Williams, 2008), exclusion (ILO, 2015b).	Duality-benign (Tokman, 1978), dualist (Chen, 2012), complement (Williams, 2008), exclusion (ILO, 2015b), alternative-neoliberal (Williams, 2008).	Subordination due to exploitation (Tokman, 1978), by-product (Williams, 2008), political economy (Williams & Kedir, 2018), driven by market competition (ILO, 2015b).

TABLE 4 Conceptual framework of the “Neoliberal”, “Free-Riding” and “Two-tiered” approaches on the informal economy (IE)

	Neoliberal	Free-riding	Two-tiered approach
General view	The IE is a market-led response by entrepreneurs to excessive state regulation (as opposed to a temporary condition of excess labor supply).	The IE is a result of firms and workers who choose to operate informally after weighing the costs and benefits of informality versus formality.	Different factors drive different segments of the IE, that is, exclusion of firms and people from state benefits or to voluntary exit decisions resulting from private cost-benefit calculations.
Types of activities	“Plucky” micro-entrepreneurial activity. Informal firms are innovators and rational actors. They are similar to official ones but are kept down by excessive taxes and regulations.	Opportunistic informal firms who choose to operate informally are constrained to operate below a certain detection threshold, therefore, lack the necessary scale to produce efficiently. According to Levenson and Maloney (1998), informal firms in developing countries behave similarly to those in industrialized countries.	High level of heterogeneity in terms of rationales and demographic characteristics between informal salaried and self-employed.
Causal roots	Informal entrepreneurs represent efficient market forces that emerge as a populist reaction to over-regulation and government oppression (Williams, 2015).	The rational choice to operate informally depends on lack of enforcement, negligible economies of scale, and benefits associated with formality. According to Farrell (2004: 28) “substantial cost advantage that informal companies gain by avoiding taxes and regulations more than offsets their low productivity and small scale.”	Two main groups of informal workers with different motivations (on average): informal self-employed opt-out of formality after implicitly making a cost-benefit analysis. They choose their occupations according to their individual needs (especially their desire for flexibility and autonomy) and abilities. Informal salaried workers are not choosing to opt-out of the formal economy. For them, exclusion rather than exit from formality appears to be the driving force behind their present informal status.

(Continues)

TABLE 4 (Continued)

Neoliberal		Free-riding		Two-tiered approach	
Nature of choice	Exit-driven due to excessive state regulation	Exit-driven due to opportunistic behavior	Exclusion-driven and exit-driven		
Policy implications	Less state regulation and more free-market policies can enable/unlock the growth potential of informal entrepreneurs (particularly through the legalization of informal property rights).	Access to mechanisms that ensure property rights. To improve law enforcement because it increases the benefits of formality. To increase enforcement of government regulations and taxation because it raises the costs of informality. Bringing informal activities into the formal regulatory environment increases the tax base and reduces unfair competition to formal businesses.	No “one-size-fits-all” solution. The policies should be tailored to the diverse characteristics, circumstances, and needs of the workers and economic units concerned. Perry et al. (2017) suggest actions to: increase the aggregate productivity in the economy; raise human capital levels (especially for the poor); reduce the costs and increase the benefits of formalization; enhance enforcement of laws and regulations; improve the quality and fairness of state institutions and policies.		
Link to the formal economy	In the short run, the IE consists of firms “exit” from the formal economy (i.e., negative correlation). In the long run, it may have a positive effect if the state reduces overburden to promote formalization.	In the short run, the IE consists of firms “exit” from the formal economy (i.e., negative correlation). In the long run, this negative effect is reinforced because informal activities are detrimental to the overall economic growth because this unfair competition reduces market efficiency.	Undetermined. It depends on the composition of the IE (e.g., the relative size of informal salaried workers and self-employed).		
Prevalent in	Developed countries	Developed countries	Coexist with different intensity in developing and developed countries.		
Proposer(s) of the theory	de Soto (1989).	Levenson and Maloney (1998), Maloney (2004), Farrell (2004), Baily et al. (2005).	Fields (1990), Feige (1997), Jacob (2003), Perry et al. (2007).		
Different labels used for comparable theoretical view	Complementarity-Benign ^a (Tokman, 1978), Legalist (Chen, 2012), Romantic (La Porta & Schleifer, 2008), Exit (ILO, 2015b).	Voluntarist (Chen, 2012), Parasite (La Porta and Schleifer, 2008), Exit (ILO, 2015b).	Heterogeneous subordination (Tokman, 1978), Holistic, Multi-segmented, Exit-Exclusion (Chen, 2012); Post-structuralist (Williams & Nadir, 2011); Institutional (Williams & Kedir, 2018); Two-tiered informality (ILO, 2015b).		

^aNote: The inclusion of Tokman's (1978) “complementarity sub-approach” together with the “neoliberal” view inspired by de Soto (1989), is not consistent with the authentic authors' views. Indeed, if for both Tokman (1978) and de Soto (1989), informality may be seen as beneficial for overall economic development and (at least, in the long-run) as complementary phenomena, the rationales behind their hypotheses are different.

row of Tables 3 and 4 (“*Different labels used for (comparable) theoretical view*”) assists the reader to match the different labels used in the literature (e.g., in Section 3.3) to the six proposed theoretical approaches.

In the last column of Table 4 – labeled as “*Two-tiered approach to informality*” following ILO (2015b) – I include all those theories³³ that, despite having different (sometimes contrasting) hypotheses on the nature and causes of IE, they share the same key idea: informal activities are so heterogeneous in motivations, characteristics, and relationship to the formal economy that a one-dimensional explanation of this phenomenon is an unsuitable approach. This theoretical approach, inspired by Fields (1990) and Perry et al. (2007), is the most utilized conceptual model for policy purposes due to its flexibility and comprehensiveness.³⁴

4 | A THEORETICAL FRAMEWORK OF INDIVIDUAL CHOICE TO OPERATE INFORMALLY

In this section, I propose a theoretical framework of individual choice to operate informally by combining the three main approaches of informality. Following the “neoclassical approach,” I define an analytical framework where the IE emerges as a voluntary choice of rational agents who maximize their welfare (utility or profit) function. According to the “macro-econometric approach,” I consider a broad set of institutional and socio-economic variables to outline the costs and benefits of the status of informal, formal, or a not-productive unit. Although the model assumes a voluntary choice, following some of the theories of the informal sectors surveyed in Section 3.3 (e.g., “Modernization,” “Dualist,” “Structuralist”), the proposed theoretical framework does not exclude that the participation in the IE may be driven by necessity (e.g., lack of employment opportunities in the formal economy, insufficient public subsidies for poor and unemployed).

The agent’s decision rule includes the variables that, according to the literature, may affect his/her welfare function in three alternative productive status: formal, informal, and nonproductive activity.

The proposed framework aims to be sufficiently flexible to account for a decision to operate informally both as a result of a “voluntary choice” of opportunistic firms/workers aiming to escape from taxation and regulatory burden, and a “forced option” for poor and marginalized people living in a deprived socio-economic context.

In general, I assume that a productive unit chooses to operate out of formal arrangements if his/her welfare the (profit/utility) function is associated with a net benefit of being a formal operator ($W_i^F = B_i^F - C_i^F$) exceeding (or equalizing to) the expected net benefit of being both informal ($W_i^{IF} = B_i^{IF} - C_i^{IF}$) and unemployed or inactive ($W_i^{UI} = B_i^{UI} - C_i^{UI}$). Welfare functions linked to these three statuses depend on a combination of demographic, institutional, and socio-economic factors.

The proposed model follows the “*Two-tiered approach*” to the IE as inspired by Fields (1990) and Perry et al. (2007) by assuming that IE depends on multiple causes and it involves heterogeneous agents in terms of characteristics and needs. In analytical terms, the decision rule consists in maximizing the welfare function of i th economic unit in three possible status: formal, informal,

and unemployed-inactive.³⁵

$$\frac{\text{Welfare function in the Formal Economy}}{W_i^F(B_i^F[Y_i^F[Sk_i, D_c], SocIns_c, LegP_c, Fm_c, AccSk_i] - C_i^F[TaxB_c, RegB_c])} \geq \frac{W_i^{IF}(B_i^{IF}[Y_i^{IF}[Sk_i, D_c], SocAss_c, (1-p)(C_i^F), AccSk_i] - C_i^{IF}[LegP_c, Fm_c, p(Sanct_c), RepC_c])}{\text{welfare function in the Informal Economy}} \geq \frac{W_i^{UI}(B_i^{UI}[SocAss_c, LeisureT_i] - C_i^{UI}[\alpha(RepC_c)])}{\text{welfare function for unemployed or inactive people}} \quad (1)$$

Specifically, as the formal operator concerns, agent i th receives the following “benefits” (B_i^F):

Y_i^F earnings of productive activity. These profits/wages depend on both individual (e.g., skills, education, socio-demographic) characteristics of enterprise/worker (Sk_i) with $\partial Y_i^F / \partial Sk_i > 0$ and macroeconomic factors related to aggregate demand of formal economy in country c th (D_c – e.g., due to business cycle fluctuations, labor market, agent’s socio-economic context³⁶ – with $\partial Y_i^F / \partial D_c > 0$). Benefits in formal economy also come from the access to public services accessible only to formal activities such as: the social welfare system ($SocIns_c$), for example, social protection schemes based on employment-related benefits, membership with trade union associations, access to better health insurance; the judiciary system for legal protection and enforceable commercial contracts ($LegP_c$); the financial system and public subsidies for business investment (Fm_c). Moreover, work experiences in the formal economy increase human capital and people’s skills that, in turn, positively affect the long-run profitability and employability of the economic units ($AccSk_i$).

The main costs to operate formally (C_i^F) consist of tax and social security burden ($TaxB_c$) and administrative costs to comply with regulations ($RegB_c$).

As the welfare function of informal operator concerns, the benefits of being informal consist of earnings/profits for units operating in the IE (Y_i^{IF}). Alike formal units, these earnings depend on both individual characteristics of productive units ($\partial Y_i^{IF} / \partial Sk_i > 0$) and aggregate demand (e.g., employment or business opportunities within the (formal and informal) economy $\partial Y_i^{IF} / \partial D_c \lesseqgtr 0$).³⁷ A second benefit of being informal, with respect to the formal activity, is that informal units may also illegitimately benefit from cash transfers, in-kind provisions, subsidies, (non-contributory) pensions, and public works programs. Accordingly, the more generous are social assistance programs ($SocAss_c$) targeted to unemployment status or poverty based on declared income, the strongest is the incentive to remain informal, other things being equal. A third benefit of being informal is that these units may produce goods or services more cheaply than their (formal) competitors by evading taxes, social contributions ($-TaxB_c$) and administrative costs to comply with regulations ($-RegB_c$). However, this competitive advantage is uncertain because it depends on the probability of being detected (p), that is, they have to pay evaded taxes and contributions to regularize their status. Consequently, informal competitors enjoy relative cost advantages that allow them to undervalue their products provided that they are not discovered, that is, $(1-p)(C_i^F)$. A fourth benefit of being informal is that, likewise formal units, also informal activities contribute to improving job opportunities and business prospects by increasing know-how and building business networks of economic units ($AccSk_i$).

I outline three main types of costs to being informal (C_i^{IF}): the (opportunity) costs related to the constraints to access to public services for individuals and firms operating in the IE. These costs are proportional to the quality and quantity of public services and constitute competitive disadvantages with respect to formal productive units $C_i^{IF} = B_i^F(LegP_c, Fm_c)$ (i.e., these costs increase as better the judiciary system for legal protection and enforceable commercial contracts and financial system and public subsidies for business investment are). A second (expected) mon-

etary cost of being informal is that if these activities are detected by the enforcement system, they have to pay administrative sanctions, that is, $C_i^{IF} = f(p(\text{Sanct}_c))$. As the third cost of informal status concerns, I include psychic and reputational costs ($\text{Rep}C_c$) usually associated with deviant behavior. In particular, I assume that (nonpecuniary) reputational costs depend on socio-cultural factors of people's (e.g., gender, age, social status) or enterprise's (e.g., size and type of business activity) context.

This theoretical framework includes a third possibility for the economic agent, he/she may voluntarily choose not to operate either in the formal or IE, that is, being inactive or unemployed. In this case, the agent's welfare function W_i^{UI} depends on the generosity of the social assistance programs and free time from child care and housework $B_i^{UI}[\text{SocAss}_c, \text{Leisure}T_i]$. An additional factor that may assume both positive (i.e., included in the B_i^{UI}) and negative effects (i.e., C_i^{UI}) on the agent's welfare is the social evaluation (α) of being inactive or unemployed. In particular, by using reputational costs ($\text{Rep}C_c$) of informal status as a benchmark, I define this factor as $\alpha(\text{Rep}C_c)$. According to this parametrization, this component represents a cost for being inactive, as in [1], if $\alpha \geq 0$ ³⁸ while it increases the welfare if $\alpha < 0$.³⁹

Consistent with this theoretical framework, also the decision to work informally as a survival strategy is a "voluntary" rational choice. Indeed, let us assume that; (a) there are no chances to operate formally because of barriers of entry (e.g., due to financial constraints, the status of an illegal immigrant, minor) or a lack of capacity of the labor market to absorb the labor force, (i.e., $W_i^F = 0$)⁴⁰; (b) informal workers receive the social subsidy targeted to (formal) unemployed and inactive people (i.e., $B_i^{IF}[\text{SocAss}_c] = B_i^{UI}[\text{SocAss}_c]$). As a result, any economic agent rationally accepts to operate informally if being informal has larger benefits than being (formally) unemployed or inactive people. In analytic terms, the decision rule to operate informally becomes:

$$Y_i^{IF}[\text{Sk}_i, D_c] + B_i^{IF}[\text{AccSk}_i, (1-p)(C_i^F)] > C_i^{IF}[\text{Leg}P_c, Fm_c, p(\text{Sanct}_c), (\alpha-1)\text{Rep}C_c] \quad (2)$$

This theoretical framework explains the larger size of the IE in developing countries compared to developed economies by combining individual and macroeconomic causes.

As unemployed and inactive people concern, in developing countries, we usually observe: (a) negligible social assistance programs due to lack of public resources ($\text{SocAss}_c \approx 0$); (b) insignificant psychic, moral, and social stigma costs ($\text{Rep}C_c \approx 0$). Indeed, in these countries, people tend to justify their participation in the IE by a lack of choice (e.g., Pardo, 1995; Gërxhani, 2004b). In conclusion, in developing countries, the benefits for nonproductive units only consists of their leisure time: $W_i^{UI} = B_i^{UI}[\text{Leisure}T_i]$.

As informal units concern, developing countries provide: (a) low-quality social protection schemes based on employment-related benefits, judiciary, financial systems, and public subsidies for (formal) business investment (i.e., $B_i^F[\text{SocIns}_c \approx 0; \text{Leg}P_c \approx 0, Fm_c \approx 0]$); (b) the survival nature of informality leads to "legitimize" deviant behavior from a social perspective (i.e., $\text{Rep}C_c \approx 0$); (c) ineffective enforcement policies to fight these activities by sanctions (e.g., because informal operators do not have sufficient economic resources to pay them or alternative source of income to survive (i.e., $p(\text{Sanct}_c) \approx 0$); (d) negligible social assistance programs due to lack of public resources ($\text{SocAss}_c \approx 0$). According to this scenario, the choice among formal, informal, or unemployed-inactive status in a developing country may be represented by

$$W_i^F(B_i^F[Y_i^F] - C_i^F[\text{Tax}B_c, \text{Reg}B_c]) \geq W_i^{IF}(B_i^{IF}[Y_i^{IF}]) \geq W_i^{UI}(\text{Leisure}T_c) \quad (3)$$

This decision rule may be further simplified in developing countries by considering two additional reasonable assumptions: (e) leisure time is a normal good – that is, the demand for it increases as income increases, therefore if you are “poor,” being inactive or unemployed is a forced status (i.e., $U_i^{UI}(LeisureT_i) \approx 0$); (f) markets are competitive therefore earnings (Y_i) only depend on units’ productivities, then agent’s choice to operate formally can be further simplified as

$$\Delta Y_i^{F-IF} \geq C_i^F [TaxB_c, RegB_c] \quad (4)$$

Where $\Delta Y_i^{F-IF} = (Y_i^F - Y_i^{IF})$ indicates the gap between formal and informal earnings.

Equation (4) allows to explain the (apparent) paradox of the “Neoliberal theory” for large size of informality in countries with limited taxation and regulatory burden (i.e., $C_i^F [TaxB_c, RegB_c] \rightarrow 0^+$). Specifically, the decision rule (Equation (4)) typically does not hold (i.e., unit opts to operate informally) because for the predominant sectors in developing economies (e.g., agriculture, retail, construction, and services) the difference in productivity between informal and formal firms is negligible due to the low optimal level of organization and limited physical capital (i.e., their remunerations are comparable $\Delta Y_i^{F-IF} \approx 0$).⁴¹

In conclusion, the proposed theoretical model, following the neoclassical approach, proposes a simplified analytical framework that deals with the issue of informality as a consequence of rational choices for welfare-maximizing agents. In line with the macro-econometric approach, it takes into account several institutional, social, cultural, macroeconomic factors as well as individual characteristics of (potentially) productive units. According to the “Two-tiered approach” to informality, it aims to be sufficiently flexible to be applied for policy purposes in both developing and developed economies.

5 | CONCLUSION

This article surveys the main definitional and theoretical approaches to informality proposed in the literature. At the state of art, we do not have a definition for “IE” that meets with both statistical standards of consistency, measurability, harmonization, and economic requirements to outline a meaningful concept of informality suitable for positive and normative economic analyses. Following Dell’Anno (2007), we can amend the ISTAT analytical framework of the NOE to obtain a definition for the IE consistent with both statistical and economic purposes. This definition includes two, of the four, components of the NOE, that is, the *underground production for economic reasons* and the *informal production*, while it excludes phenomena such as *underground production for statistical reasons* and *illegal production* that are not related to the usual aims pursued by the economic analysis of informality. Moreover, this definition attempts to meet the measurement purposes with the economic research that points out some structural differences between “underground” economy and “informal” production. Indeed, in line with the *Two-tiered* approach to informality, while *informal production* (i.e., where unregistered units with little or no division between labor and capital and on a small-scale as casual self-employment and informal salaried workers operate) is a typical consequence of “exclusion,” *underground production for economic reasons* (e.g., where self-employed, freelancers, second workers, business activities that deliberately hide their earnings to reduce tax or regulatory burden operate), is a usual (but not exclusive) result of a voluntary “exit” choice. The proposed definition may also be sufficiently flexible for cross-countries analyses. For instance, we expect to find a predominant share of *under-*

ground production for economic reasons in developed countries while the relative size of *informal production* is expected to be prevalent in developing countries.

The second aim of this research is to survey the literature on the theories of informality. I distinguish three, not mutually exclusive, approaches: the neoclassical, the macro-econometric, and the conceptual perspectives. Given the partially overlapping classifications and theories, I propose a theoretical framework able to incorporate the main schools of thought on IE by outlining six main theoretical views.

The survey concludes by proposing a unified analytical framework where the multifaceted phenomenon of IE emerges as a result of the agent's rational choice.

From a normative viewpoint, I have highlighted as different schools of thought endorse diverse policy measures. However, according to the prevalent literature, the most useful approach for policy purposes should focus on two main dimensions: the nature of the choice (voluntary/forced) to operate informally (i.e., Perry et al., 2007) and the heterogeneous features of informal workers and firms. This view, known as the *Two-tiered approach* to informality, is consistent with the evidence that several causes of informality coexist in the same economy and, within the same country, across different sectors. In particular, the literature has pointed out that qualitative differences exist in the composition of IE across developed and developing nations, with the informal determinants of developed countries different compared to those of developing economies (e.g., Goel & Nelson, 2016).⁴² The normative implication of this result is that public policies, to be effective, have to take into account both the heterogeneous nature of informal firms and agents and their relative size in the national economy. In conclusion, I agree with the ILO's (2015) policy guidelines, that it is wrong to look for a "*one-size-fits-all*" policy response to the IE valid across countries or even across different components of the IE within a country. The design of policy interventions depends on the understanding of the people's reasons to work informally in a given context. For that reason, the assessment of the socio-economic characteristics and needs of activities that choose to operate informally, compared to those in which the informality is a result of the "exclusion" from the formal sector, becomes pivotal for the success of whatever formalization strategy.

As a final point, I believe that the IE represents a research field with significant opportunities for future research. Several questions and debates remain on the table and, as this article has attempted to do, only by increasing the collaboration between statisticians and economists, we could find both a definition of informality and a general theory of the IE suitable for explanatory, policy, and measurement purposes.

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ENDNOTES

¹ Precisely, ILO (2018a, p.15) reports that in emerging and developing countries, the share of informal employment is 69.6% of the employed population while in developed countries, the informal employment is equal to 18.3%.

² See Section 2 for a discussion of the different meanings of the concepts of "informal sector" and "IE."

- ³ For the *measurement issue*, OECD (2002) distinguishes two main approaches to estimate the IE: the econometric methods and the National Accounting Approach. The former includes the “direct methods” (i.e., based on direct information about informal activities as tax returns, administrative data, questionnaire surveys, or experimental data), the “indirect methods,” that is, measuring the “traces” that the IE leaves in official macroeconomic data and the Multiple Indicators and Multiple Causes approach, that is, based on the statistical approach of the Structural Equation Modeling. See for a review of this literature: Schneider and Enste (2000), OECD (2002), Schneider (2005), ILO (2013), Dell’Anno and Davidescu (2019), Elgin and Erturk (2019), Bashlakova and Bashlakov (2021).
- ⁴ Specifically, the statistical manual to measuring informality of ILO (2013, p.44) reports that “while other ILO publications and some documents published by other organizations use the term ‘IE,’ this manual prefers to use the terms “informal sector” and “informal employment.”
- ⁵ According to the 17th ICLS (ILO, 2003, p.5), informal employment includes “the total number of informal jobs, whether carried out in formal sector enterprises, informal sector enterprises, or households, during a given reference period.”
- ⁶ It is important to point out that ILO’s splitting of productive activities between informal and formal sectors does not perfectly match with the SNA. According to SNA (2008, p.477) units that are not informal are not necessarily defined as formal in the SNA. This occurs for *households with unincorporated enterprises not included in the informal sector that are divided between those that are treated as formal (because of size or registration) and the rest that are not treated as informal but are left simply in a group called households.* Further difference exists on ILO concepts of “sector” and “market production” that do not conform to the SNA. As the latter concerns, “the ILO treats an enterprise as a market producer if any of the output is sold whereas the SNA requires that most or all of the output be sold.” (SNA 2008, p.478)
- ⁷ Here the term “illicit” stands for “criminal activities.” Therefore the provision of services or the production, sale, possession or use of goods forbidden by law (e.g., production and trafficking of drugs, firearms trafficking) are excluded but noncompliance behaviors, such as selling in a street without a license, are included in the IE.
- ⁸ According to ESA (2010), the NOE encompasses “value of production activities that are not directly observed are, in principle, included within the national accounts production boundary.”
- ⁹ The reasons to conceal “shadow” productions are inspired by SNA (2008, par. 6.40).
- ¹⁰ Indeed, in this case, these activities are not productive because they only modify the income distribution between taxpayers and the public sector.
- ¹¹ Specifically, OECD (2002, p.13) defines underground production for economic reasons as “those activities that are productive and legal but are deliberately concealed from the public authorities to avoid payment of taxes or complying with regulations.”
- ¹² This resolution will be presented for consideration at the 21st ICLS, which is expected to take place in 2023.
- ¹³ In particular, the IME does not include those IPA that are carried out for an economic unit that is not producing for the market for pay or profit – such as own-use production work, volunteer work, or unpaid trainee work – nor illegal production – that is, illegal activities where the parties are willing partners in an economic transaction forbidden by law or that are unlawful when carried out by unauthorized units – or under declared market value transactions that affect the amount of value added of market transactions.
- ¹⁴ *Economic underground due to underreporting*, that is, production/value added may be understated in order to avoid taxes, social charges, and so forth.
- ¹⁵ *Economic underground due to not registered units*, that is, enterprises may be missing because the owners have deliberately avoided the obligations to register in order to avoid additional costs of various kinds such as value added taxes, social security contributions, costs related to the compliance with health and safety standards, and so forth.
- ¹⁶ *Informal sector: units not registered*, that is, non-registration can be a criterion for defining the informal sector and enterprises may be missing simply because they are not required to register by any kind of legislation.
- ¹⁷ As the effects of the IE concern, see Elgin and Erturk (2019) for a recent survey on the literature that analyzes the consequences of IE on the economic system.
- ¹⁸ Gerry (1987) provides an interesting contextualization of the nature and role of the IE as closely connected to the theories of socio-economic development. For the sake of brevity, we do not report this classification.

- ¹⁹ For example, Lewis's (1954) dual-sector model; Rosenstein-Rodan's (1943) external economies and "big push"; Rostow's (1960) "stages of economic growth." Meier (2000) defines this "school" as the first generation of development economists that considered the state as the major agent of change.
- ²⁰ For instance, ILO (1991, p.63) explicates the novelty of this new perspective on the informal sector arguing that: "Contrary to earlier beliefs, the informal sector is not going to disappear spontaneously with economic growth. It is, on the contrary, likely to grow in the years to come, and with it the problems of urban poverty and congestion will also grow." Hart (1973, pp.88–89) concluded his analysis by suggesting more empirical investigation to determine which theory best explains the persistence of the informal sector. He considered two alternative perspectives: the "Socialists [who] argue that foreign capitalist dominance [...] condemns the majority of the urban population to deprivation and exploitation" and "More optimistic liberals [who see] in informal activities [...] the possibility of a dramatic "bootstrap" operation, lifting the underdeveloped economies through their own indigenous enterprise."
- ²¹ For example, ILO (1972).
- ²² For example, McGee (1973).
- ²³ A link between Hirschman's (1970) "exit" and "voice" options and the individual's decision to work informally has also been proposed by Schneider and Enste (2000). However, as Perry et al. (2007) point out, Schneider and Enste's emphasis was on the reaction of individuals who feel overburdened by the state, while Perry et al. deal with the perceived inadequacy of the benefits that formality entails.
- ²⁴ Williams's analysis refers to "informal work" rather than "IE." However, his interpretations can be generalized to the IE.
- ²⁵ He outlined four perspectives, informality "as a residue," "as a by-product," "as a complement," and "as an alternative" to the formal economy.
- ²⁶ Williams and his co-authors use different labels for this theory, that is, "as a by-product" in Williams (2008), *Structuralist theory* in Williams and Nadin (2011) and *Political economy theory* in Williams (2015) and Williams and Kedir (2018). Other scholars label this approach as *Neo-Marxist* (e.g., Biles, 2009).
- ²⁷ Williams (2008) includes this theory as one of the "alternative approaches" to informality, specifically, the "neoliberalism" view.
- ²⁸ See Williams and Nadin (2011) for this literature.
- ²⁹ It corresponds to the "Parasite" view in La Porta and Shleifer (2008).
- ³⁰ It corresponds to the "Romantic" view in La Porta and Shleifer (2008).
- ³¹ The idea of a conceptual framework follows Brown and McGranahan (2016). They provide a similar table to summarize Chen's (2012) classification of the theories on the IE.
- ³² If the sign of the relationship between formal and IE is not clearly indicated by the schools of thought, we infer the "school" by the authors' arguments. For a general discussion on this topic, see, among the others, Gërzhani (2004b), Schneider (2005), Dell'Anno (2008), and Elgin et al. (2021).
- ³³ For example, *Heterogeneous subordination* proposed by Tokman (1978); *Holistic, Multi-segmented, Exit-Exclusion* (Chen, 2012); *Post-structuralist* (Williams and Nadir, 2011); *Institutional* (Williams and Kedir 2018); *Two-tiered informality* proposed by ILO (2015b) and inspired to Fields (1990) and Perry et al. (2007).
- ³⁴ Indeed, ILO's (2015b) guide for national employment policies applies this theoretical ground for policy responses to the IE.
- ³⁵ We could also include agent's welfare associated with the choice to be a criminal by adding $(Y_i^{Cr} + B_i^{Cr}[SocAss_c] - C_i^{Cr}[\beta(CrimSanct_c), \gamma(RepC_c)])$ where: β denotes the probability of arrest and conviction for criminal activity in the country c ; $CrimSanct_c$ the severity of the sentence; $RepC_c$ accounts for social stigma and psychic costs, and $\gamma > \alpha > 1$ (see footnote 38 for details on α). However, we exclude this status from the theoretical framework because out of the scope of this research.
- ³⁶ With "socio-economic context" we mean that the aggregate demand that faces the productive unit may depend on several factors, such as the degree of economic development (e.g., developing vs. developed economy), the geographic area (rural vs. urban).
- ³⁷ The sign of the relationship between formal and IE is debatable and it depends on the business cycle as well. See Elgin et al. (2021) for an empirical analysis of this topic.
- ³⁸ Specifically, $\alpha = 0$ means no social stigma and psychic costs for inactive and unemployed, $\alpha = 1$ indicates that this type of nonmonetary costs is equal between informal operators and unemployed or inactive people and $\alpha > 1$ implies that for the agent's social context being inactive or unemployed is worse than operating in the IE in terms of social stigma.

- ³⁹ For example, this may be the case of disapproving wisdom towards woman participation in the (formal or informal) labor force present in some cultures.
- ⁴⁰ For example, because of a lack of human and/or physical capital to enter in the formal economy. This lack of the agent's endowment may be due to both "micro" characteristics as inadequate education achievement (Sk_i) and macro factors D_c (e.g., related to economic recession, sectoral change in production, deindustrialization)
- ⁴¹ The standard hypothesis is that the informal unit productivity is lower than the formal one because the former has lower human and physical capital endowments, restricted access to public services, or other benefits from regulation than formal units (Loayza & Rigolini 2011). Consequently, for those activities with a fairly small optimal levels of inputs endowment and/or where the access to public services is unimportant, operating formally does not provide effective advantages in terms of efficiency compared to informal ones.
- ⁴² For instance, while factors like inflation and tax complexity promote informal activities in both cases, other regulatory barriers (e.g., business startup procedures) incentive informality only in developing countries.

REFERENCES

- Afonso, O., Neves, P. C., & Pinto, T. (2020). The non-observed economy and economic growth: A meta-analysis. *Economic Systems*, 44(1), 100746.
- Allen, J., Nataraj, S., & Schipper, T. C. (2018). Strict duality and overlapping productivity distributions between formal and informal firms. *Journal of Development Economics*, 135, 534–554.
- Allingham, M. G., & Sandmo, A. (1972). Income tax evasion: A theoretical analysis. *Journal of Public Economics*, 1(3–4), 323–338.
- Alm, J. (2019). What motivates tax compliance? *Journal of Economic Surveys*, 33, 353–388.
- Amaral, P. S., & Quintin, E. (2006). A competitive model of the informal sector. *Journal of Monetary Economics*, 53(7), 1541–1553.
- Baily, M., Farrell, D., & Remes, J. (2005). *Domestic services: The hidden key to growth*. McKinsey Global Institute.
- Bangasser, P. E. (2000). The ILO and the informal sector: An institutional history [Employment Paper 2000/9]. Employment Sector, International Labour Office.
- Bashlakova, V., & Bashlakov, H. (2021). The study of the shadow economy in modern conditions: Theory, methodology, practice. *The Quarterly Review of Economics and Finance*, 81, 468–480.
- Biles, J. J. (2009). Informal work in Latin America: Competing perspectives and recent debates. *Geography Compass*, 3(1), 214–236.
- Bonnet, F., & Venkatesh, S. (2016). Poverty and informal economies. In D. Brady & L. Burton (Eds.), *Oxford handbook of the social science of poverty* (pp. 637–659), Oxford University Press.
- Brown, D., & McGranahan, G. (2016). The urban informal economy, local inclusion and achieving a global green transformation. *Habitat International*, 54, 97–105.
- Cagan, P. (1958). The demand for currency relative to the total money supply. *Journal of Political Economy*, 66(3), 302–28.
- Calzaroni, M. (2000). The exhaustiveness of production estimates: New concepts and methodologies. In *Proceedings of the International Conference on Establishment Surveys*. Buffalo, Statistics Canada.
- Capasso, S., & Jappelli, T. (2013). Financial development and the underground economy. *Journal of Development Economics*, 101, 167–178.
- Castells, M., & Portes, A. (1989). World underneath: The origins, dynamics and effects of the informal economy. In A. Portes, M. Castells, & L. Benton (Eds.), *The informal economy: Studies in advanced and less developed countries*. Johns Hopkins University Press.
- Charmes, J. (2016). *The informal economy: What is it? Where does it come from? How big is it? Why is it growing? How to tackle it? Research, network and support facility, volume 2 – definition of the informal economy*. ARS Progetti.
- Chen M. & Carré F. (Eds.). (2020). *the informal economy revisited*. Routledge.
- Chen, M., Vanek, J., & Carr, M. (2004). *Mainstreaming informal employment and gender in poverty reduction: A handbook for policymakers and other stakeholders*. Commonwealth Secretariat.
- Chen, M. A. (2012). The informal economy: Definitions, theories and policies [WIEGO Working Paper No. 1]. WIEGO.
- Chong, A., & Gradstein, M. (2007). Inequality and informality. *Journal of Public Economics*, 91(1–2), 159–179.

- Clement, C. (2015). *The formal-informal economy dualism in a retrospective of economic thought since the 1940s*, *schriftenreihe des promotionsschwerpunkts globalisierung und beschäftigung*, No. 43/2015, Universität Hohenheim.
- Davis, M. (2006). *Planet of slums*. Verso.
- de Soto, H. (1989). *The other path*. Harper and Row.
- de Soto, H. (2000). *The mystery of capital: Why capitalism triumphs in the west and fails everywhere else*. Basic Books.
- Dell'Anno, R. (2007). The shadow economy in Portugal: An analysis with the MIMIC approach. *Journal of Applied Economics*, 10(2), 253–277.
- Dell'Anno, R. (2008). What is the relationship between unofficial and official economy? An analysis in Latin American countries. *European Journal of Economics, Finance and Administrative Sciences*, 12, 185–203.
- Dell'Anno, R. (2016). Analyzing the determinants of the shadow economy with a 'separate approach'. An application of the relationship between inequality and the shadow economy. *World Development*, 84, 342–356.
- Dell'Anno, R. (2018). Inequality, informality, and credit market imperfections. *Macroeconomic Dynamics*, 22(5), 1184–1206.
- Dell'Anno, R. (2021a). Theoretical approaches to the phenomenon of informality. In S. M. Ruesga, J. Baquero, & J. L. Delgado (Eds.), *Dialogues on Socioeconomy: Informality in Latin America* (chapter 3) (pp. 117–134). Tirant lo Blanc.
- Dell'Anno, R. (2021b). Inequality and informality in transition and emerging countries. *IZA World of Labor*, 325, <https://doi.org/10.15185/izawol.325.v2>
- Dell'Anno, R., & Davidescu, A. (2019). Estimating shadow economy and tax evasion in Romania. A comparison by different estimation approaches. *Economic Analysis and Policy*, 63(C), 130–149.
- Devos, K. (2014). Tax compliance theory and the literature. In *Factors influencing individual taxpayer compliance behaviour*. Springer.
- Dreher, A., & Schneider, F. (2010). Corruption and the shadow economy: An empirical analysis. *Public Choice*, 144(1–2), 215–238.
- Elgin, C., & Erturk, F. (2019). Informal economies around the world: Measures, determinants and consequences. *Eurasian Economic Review*, 9(2), 221–237.
- Elgin, C., Kose, M. A., Ohnsorge, F., & Yu, S. (2021). Growing apart or moving together? Synchronization of informal and formal economy business cycles. In F. Ohnsorge & S. Yu (Eds.), *The long shadow of informality: Challenges and policies*. World Bank.
- ESA (2010). *European system of national and regional accounts 2010*. Publications Office of the European Union. 2013.
- Farrell, D. (2004). The hidden dangers of the informal economy. *McKinsey Quarterly*, 3, 26–37.
- Feige, E. L. (1979). How big is the irregular economy? *Challenge*, 22(1), 5–13.
- Feige, E. L. (1997). Underground activity and institutional change: Productive, protective and predatory behavior in transition economies. In J. M. Nelson, C. Tilley, & L. Walker (Eds.), *Transforming post-communist political economies* (pp. 21–35). National Academy Press.
- Feige, E. L. (2016). Reflections on the meaning and measurement of unobserved economies: What do we really know about the "Shadow Economy"? *Journal of Tax Administration*, 2(1), 5–41.
- Feld, L. P., & Schneider, F. (2010). Survey on the shadow economy and undeclared earnings in OECD countries. *German Economic Review*, 11(2), 109–149.
- Ferreira-Tiryaki, G. (2008). The informal economy and business cycles. *Journal of Applied Economics*, 11, 91–117.
- Fields, G. S. (1990). Labor market modelling and the urban informal sector: Theory and evidence. In D. Turnham, B. Salomé, & A. Schwarz (Eds.), *The informal sector revisited* (pp. 49–69). OECD.
- Gaertner W. & Wenig A., (Eds.). (1985). *The economics of the shadow economy*. Springer Publishing Company.
- Geertz, C. (1963). *Old societies and new states: The quest for modernity in Asia and Africa*. Free Press.
- Gerry, C. (1974). Petty producers and the urban economy: A case study of Dakar. WEP Urbanization and Employment Research Programme [Working Paper No. 8]. ILO.
- Gerry, C. (1987). Developing economies and the informal sector in historical perspective. *The Annals of the American Academy of Political and Social Science*, 493, 100–119.
- Gërkhani, K. (2004a). The informal sector in developed and less developed countries: A literature survey. *Public Choice*, 120, 267–300.

- Gërzhani, K. (2004b). Tax evasion in transition: Outcome of an institutional clash? Testing Feige's conjecture in Albania. *European Economic Review*, 48(4), 729–745.
- Godfrey, P. C. (2015). Introduction: Why the informal economy matters to management. In P. C. Godfrey (Ed.), *Management, society, and the informal economy* (pp. 1–20). Routledge.
- Goel, R. K., & Nelson, M. A. (2016). Shining a light on the shadows: Identifying robust determinants of the shadow economy. *Economic Modelling*, 58(C), 351–364.
- Goel, R. K., & Saunoris, J. W. (2014). Global corruption and the shadow economy: Spatial aspects. *Public Choice*, 161(1–2), 119–139.
- Granda-Carvajal, C. (2010). The unofficial economy and the business cycle: A test for theories. *International Economic Journal*, 24(4), 573–586.
- Günther, I., & Launov, A. (2012). Informal employment in developing countries: Opportunity or last resort? *Journal of Development Economics*, 97(1), 88–98.
- Gutmann, P. M. (1977). The subterranean economy. *Financial Analysts Journal*, 33(6), 26–27.
- Harris, J. R., & Todaro, M. P. (1970). Migration, unemployment and development: A two-sector analysis. *American Economic Review*, 60(1), 126–142.
- Harriss-White, B. (2020). India's informal economy. Past, present and future. In M. Chen & F. Carré (Eds.), *The informal economy revisited* (pp. 38–44). Routledge.
- Hart, K. (1973). Informal income opportunities and urban employment in Ghana. *The Journal of Modern African Studies*, 11(1), 61–89.
- Hirschman, A. (1970). *Exit, voice and loyalty*. Harvard University Press.
- Husmanns, R. (2004). Measuring the informal economy: From employment in the formal sector to informal employment, [Integration Working Paper No. 53]. ILO.
- Ihrig, J., & Moe, M. (2004). Lurking in the shadows: The informal sector and government policy. *Journal of Development Economics*, 73, 541–557.
- ILO (1972). *Employment, incomes and equality. A strategy for increasing productive employment in Kenya*. International Labour Office.
- ILO (1991). The dilemma of the informal sector. Report of the Director-General. In *Proceedings of the International Labour Conference, 78th Session*. International Labour Office.
- ILO (1993). Resolutions concerning international classification of status in employment. In *Proceedings of the 15th International Conference of Labour Statisticians*. January 1993.
- ILO (2003). Guidelines concerning a statistical definition of informal employment. In *Proceedings of the 17th International Conference of Labour Statisticians (17th ICLS Guidelines)*, 24 November–3 December.
- ILO (2013). *Measuring informality: A statistical manual on the informal sector and informal employment*, International Labour Office.
- ILO (2015a). *R204 – Transition from the informal to the formal economy recommendation: Recommendation concerning the transition from the informal to the formal economy*. International Labour Organisation.
- ILO (2015b). *National employment policies – A guide for workers' organisations*. International Labour Office.
- ILO (2018a). *Women and men in the informal economy: A statistical picture* (3rd ed.). International Labour Office.
- ILO (2018b). Revision of the 15th ICLS resolution concerning statistics of employment in the informal sector and the 17th ICLS guidelines regarding the statistical definition of informal employment. In *Proceedings of the 20th International Conference of Labour Statisticians*, 1–19 October 2018 Room Document 17.
- ILO (2021). Conceptual framework for statistics on informal economy. Department of Statistics, Working Group for the Revision of the standards for statistics on informality. Draft under construction for discussion at the third meeting of the Working Group (31-01-2021).
- Jacob, J. (2003). Alternative lifestyle spaces. In A. Leyshon, R. Lee, & C. C. Williams (Eds.), *Alternative economic spaces*. Sage.
- Jessen, J., & Kluve, J. (2021). The effectiveness of interventions to reduce informality in low- and middle-income countries. *World Development*, 138, 105256.
- La Porta, R., & Shleifer, A. (2008). The unofficial economy and economic development. *Brookings Papers on Economic Activity*, 2008, 275–363.
- La Porta, R., & Shleifer, A. (2014). Informality and development. *The Journal of Economic Perspectives*, 28(3), 109–26.

- Lautier, B. (1990). La girafe et la licorne: du secteur informel au système d'emploi en Amérique latine [*Rapport de recherche*]. Ministère de l'Education Nationale, Informalité, formation et emploi: une comparaison entre la Colombie et le Nordeste Brésilien, rapport collectif GREITD/MEN.
- Levenson, A. R., & Maloney, W. F. (1998). The informal sector, firm dynamics, and institutional participation [*Policy Research Working Paper Series 1988*]. The World Bank.
- Lewis, A. (1954). Economic development with unlimited supplies of labour. *Manchester School of Economic and Social Studies*, 22, 139–191.
- Loayza, N. V. (2018). Informality: Why is it so widespread and how can it be reduced? World Bank Research and Policy Briefs No. 133110.
- Loayza, N. V., & Rigolini, J. (2011). Informal employment: Safety net or growth engine? *World Development*, 39(9), 1503–1515.
- Maloney, W. F. (2004). Informality revisited. *World Development*, 32, 1159–1178.
- McGee, T. G. (1973). *Hawkers in Hong Kong: A study of planning and policy in a third world city*. University of Hong Kong, Centre of Asian Studies.
- Medina, L., & Schneider, F. (2018). *Shadow economies around the world: What did we learn over the last 20 years?* [IMF Working Papers 2018/017]. International Monetary Fund.
- Meier, G. M. (2000). The old generation of development economists. In G. M. Meier & J. E. Stiglitz (Eds.), *Frontiers of development economics: The future in perspective*. World Bank and Oxford University Press.
- Mkandawire, P. T. (1985). The informal sector in the labour reserve economies of Southern Africa with special reference to Zimbabwe [*Working Paper No. 1*]. Institute of Development Studies, University of Zimbabwe.
- Moser, C. N. (1978). Informal sector or petty commodity production: Dualism or independence in urban development. *World Development*, 6, 1041–1064.
- Murphy, K. M., Shleifer, A., & Vishny, R. W. (1989). Industrialization and the Big Push. *Journal of Political Economy*, 97(5), 1003–1026.
- OECD (2002). *Measuring the non-observed economy – A handbook. organization for economic cooperation and development*. OECD Publishing.
- Ohnsorge, F., & Yu, S. (2021). *The long shadow of informality: Challenges and policies*. World Bank.
- Oviedo, A. M., Thomas, M. R., & Karakurum-Özdemir, K. (2009). Economic informality: Causes, costs, and policies: A literature survey [*Working Paper no. 16*]. The International Bank for Reconstruction and Development, The World Bank.
- Pardo, I. (1995). Morals of legitimacy in Naples: Streetwise about legality, semi- legality, and crime. *European Journal of Sociology*, 36(1), 44–71.
- Perry, G., Arias, O., Fajnzylber, P., Maloney, W., Mason, A., & Saavedra, J. (2007). *Informality: Exit and exclusion*. World Bank.
- Portes, A. (1994). The informal economy and its paradoxes. In N. J. Smelser & R. Swedberg (Eds.), *The handbook of economic sociology*. Princeton University Press.
- Quijano, A. (1974). The marginal pole of the economy and the marginalized labour force. *Economy and Society*, 3(4), 393–428.
- Quiros-Romero, G., Alexander, T. F., & Ribarsky, J. (2021). Measuring the informal economy [*IMF Policy Paper No. 2021/002*].
- Rauch, J. (1991). Modeling the informal sector formally. *Journal of Development Economics*, 35(1), 33–47.
- Rosenstein-Rodan, P. (1943). Problems of Industrialization of Eastern and Southeastern Europe. *Economic Journal*, 53(210/211), 202–211.
- Rostow, W. (1960). *The stages of economic growth: A non-communist Manifesto* (1st ed.). Cambridge University Press.
- Schneider, F., & Williams, C. C. (2013). *The shadow economy*. Institute of Economic Affairs.
- Schneider, F. (2005). Shadow economies around the world: What do we really know? *European Journal of Political Economy*, 21, 598–642.
- Schneider, F. (2011). *Handbook on the shadow economy*. Edward Elgar Publishing.
- Schneider, F. (2019). Shadow economy. In A. Marciano & G. B. Ramello (Eds.), *Encyclopedia of law and economics*. Springer.
- Schneider, F., & Buehn, A. (2017). Estimating a shadow economy: Results, methods, problems, and open questions, De Gruyter open. *Open Economics*, 1, 1–29.

- Schneider, F., & Enste, D. H. (2000). Shadow economies: size, causes, and consequences. *Journal of Economic Literature*, 38(1), 77–114.
- SNA (2008). *System of National Accounts 2008*. United Nations, 2009.
- Tanzi, V. (1980). The underground economy in the United States: Estimates and implications. *Banca Nazionale del Lavoro*, 135(4), 427–453.
- Thomas, J. J. (1995). *Surviving in the city: The urban informal sector in Latin America*. PlutoPress.
- Tokman, V. E. (1978). An exploration into the nature of informal-formal sector relationships. *World Development*, 6(9–10), 1065–1075.
- Ulyssea, G. (2018). Firms, informality, and development: Theory and evidence from Brazil. *American Economic Review*, 108(8), 2015–2047.
- Ulyssea, G. (2020). Informality: Causes and consequences for development. *Annual Review of Economics*, 12(1), 525–546.
- Webb, J. W., Ireland, R. D., & Ketchen, D. J. (2014). Toward a greater understanding of entrepreneurship and strategy in the informal economy. *Strategic Entrepreneurship Journal*, 8, 1–15.
- Whitson, R. (2007). Hidden struggles: Spaces of power and resistance in informal work in urban Argentina. *Environment and Planning A*, 39(12), 2916–2934.
- Williams, C. C. (2015). Out of the shadows: Classifying economies by the extent and nature of employment in the informal economy. *International Labour Review*, 154, 331–351.
- Williams, C. C., & Kedir, A. (2018). Evaluating competing theories of informal sector entrepreneurship: A study of the determinants of cross-country variations in enterprises starting-up unregistered. *The International Journal of Entrepreneurship and Innovation*, 19(3), 155–165.
- Williams, C. C. (2008). A critical evaluation of competing representations of the relationship between formal and informal work. *Community, Work & Family*, 11(1), 105–124.
- Williams, C. C., & Nadin, S. (2011). Theorising the hidden enterprise culture: The nature of entrepreneurship in the shadow economy. *International Journal of Entrepreneurship and Small Business*, 14(3), 334–348.
- Wilson, T. D. (2010). An introduction to the study of informal economies. *Urban Anthropology and Studies of Cultural Systems and World Economic Development*, 39(4), 341–357.

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